

FUNCTIONS



SYNOPSIS

➤ Function or mapping :

A relation 'f' from a set A to a set B is said to be function or mapping if every element of set A has associated with unique element in set B. It is denoted by $f : A \rightarrow B$.

➤ Image and Pre-Image: If 'f' is a function from A to B and $(a, b) \in f$, then $f(a) = b$, where 'b' is called the image of 'a' under f and 'a' is called the pre-image of 'b' under 'f'.

Note: The number of functions from A to B is $\{n(B)\}^{n(A)}$

➤ Domain, codomain, Range : Let $f : A \rightarrow B$ be a function then A is called the domain and B is called the co-domain of the function f.

➤ If $f : A \rightarrow B$ is a function, then $f(A)$, the set of all f-images of elements in A, is called the range of f

➤ The range of a function f denoted by $f(A)$, and $f(A) \subseteq B$.

➤ One-one function (Injection) :

A function $f : A \rightarrow B$ is one-one (injection) if distinct elements of A have distinct images in B.

Let $a_1, a_2 \in A$ and $a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)$

(or) $a_1, a_2 \in A$ and $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$ only.

➤ Let $n(A) = r$ and $n(B) = n$ then the condition to define an injection from A to B is $r \leq n$ and the number of such injections is ${}^n P_r$.

➤ Note: If $r > n$ then the number of injections is 0.

➤ Many-one function: (i) Let $x_1, x_2 \in$ domain of f. If $x_1 \neq x_2 \Rightarrow f(x_1) = f(x_2)$ for every x_1, x_2 in the domain, then f is one-one, other wise many-one. (ii) Consider any two points $x_1, x_2 \in X$. Take $f(x_1) = f(x_2)$ and solve the equation if we get $x_1 = x_2$ only then f is one-one

other wise it is many -one .

(iii) **Horizontal line test :** If any straight line parallel to x-axis intersects the graph of the function atmost at one point then the function is one-one, other wise it is many-one.

(i.e., intersects the graph of the function in atleast two points).

(iv) All even functions are many-one

(v) If the function is continuous and entirely strictly increasing or strictly decreasing in the domain, then f is one - one, other wise many-one .

(vi) All periodic functions are many one.

➤ Onto function (Surjection) :

$f : A \rightarrow B$ is called an onto or a surjection if every element of B has atleast one pre-image in A.

➤ If $f : A \rightarrow B$ is onto (a surjection) then the range = codomain of 'f' i.e., $f(A) = B$.

➤ The condition for a function $f : A \rightarrow B$ to be a surjection from A to B is $n(A) \geq n(B)$.

➤ If $n(A) = n(\geq 2)$, $n(B) = 2$ then the number of onto functions from A to $B = 2^n - 2$

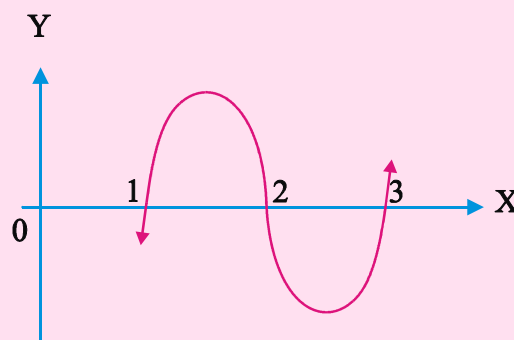
➤ If $n(A) = r$ and $n(B) = n(\geq 2)$. Then the number of surjections from A to B is $n^r - {}^n C_1(n-1)^r + {}^n C_2(n-2)^r - {}^n C_3(n-3)^r + \dots + (-1)^{n-1} \cdot {}^n C_{n-1}$.

➤ Note: If $n(A) < n(B)$ then the no. of onto functions is zero.

W.E.1: $f : R \rightarrow R$ defined by

$f(x) = (x-1)(x-2)(x-3)$ then $f(x)$ is

Sol:



Graphically, $y = f(x) = (x-1)(x-2)(x-3)$, which is clearly many - one and onto.

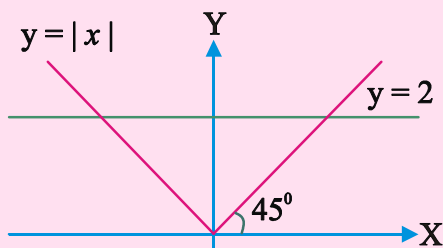
- **Into function :** If $f : A \rightarrow B$ is not onto then it is called an into function. i.e., there is no pre-image for atleast one element of B in A then f is into function.
- **Working Rule for checking whether the function $f : A \rightarrow B$ is onto or into:**

(i) Let $y \in B$ and $y = f(x)$ and from this find x in terms of y . $\forall y \in B$ if there exists atleast one $x \in A$, then f is onto. Otherwise into. i.e., if range of $f = B$, then f is onto, other wise it is into.

(ii) **Horizontal line test :** If every straight line parallel to x-axis from points in the codomain intersects the graph of the function at atleast one point then f is onto. If the line does not cut the graph of $y = f(x)$ then f is into.

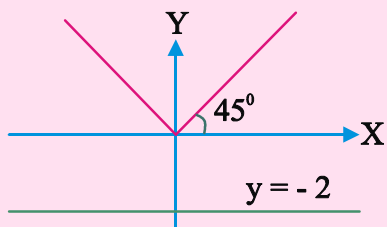
W.E.2: $f : \mathbb{R} \rightarrow [0, \infty)$, $f(x) = |x|$ is an onto function

Sol:



W.E.3: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x|$ is an into function

Sol:



- **Bijection (or) one-one & onto function:**

If $f : A \rightarrow B$ is both an injection and a surjection then f is said to be bijection or one to one and onto from A to B.

i) If A,B are finite sets and $f : A \rightarrow B$ is a

bijection then $n(A) = n(B)$.

ii) Identity function on any non empty set A is bijection.

iii) If A,B are finite sets and $n(A) = n(B)$ then number of bijective functions defined from A to B is $(n(A))!$

iv) A constant function is bijection if $n(A) = n(B) = 1$

v) **Horizontal line test:** If every straight line parallel to x-axis from the points in codomain intersects the graph of the function at only one point then f is bijection.

- **Equality of Functions:** The functions f and g are said to be equal if,

i) the domain of f = the domain of g

ii) $f(x) = g(x)$ for every x in domain.

W.E-4: State whether $f(x) = x$; $g(x) = \sqrt{x^2}$ are identical or not

Sol. $D_f = D_g = \mathbb{R}$ but $R_f = \mathbb{R}$ where $R_g = [0, \infty) \Rightarrow f(x) \neq g(x)$

(2) State where $f(x) = \operatorname{sgn}(x^2 + 1)$ and $g(x) = \sin^2 x + \cos^2 x$ are equal (or) not

Sol.

$D_f = \mathbb{R} = D_g$ and $f(x) = 1 (\because x^2 + 1 > 0)$

$g(x) = 1 \Rightarrow f(x) = g(x)$

- **Constant function:** A function $f : A \rightarrow B$ is a constant function if the range of f contains only one element.

➤ The number of constant functions from A to B is $n(B)$.

➤ The graph of constant function is a line parallel to x-axis.

➤ Range of any constant function is a singleton set.

➤ **Identity function:** Let A be a non - empty set then $f : A \rightarrow A$ defined by $f(x) = x, \forall x \in A$ is called the identity function on A and it is denoted by I_A .

➤ The graph of identity function is a straight line passing through origin and inclined at an angle of 45° with x-axis.

E.g: Let $A = \{1, 2, 3, 4\}$, then the identity function on A is $I_A = \{(1, 1)(2, 2)(3, 3)(4, 4)\}$

➤ **Composite function:** If $f: A \rightarrow B$ and $g: B \rightarrow C$ are functions then $g \circ f: A \rightarrow C$ is defined by $(g \circ f)(x) = g\{f(x)\} \forall x \in A$ is called the composite function of f & g .

➤ **Inverse function:** A function $f: X \rightarrow Y$ is defined to be invertible if there exists a function $g: Y \rightarrow X$ such that $g \circ f = I_X$ and $f \circ g = I_Y$. The function g is called the inverse of f , and is denoted by f^{-1} .

➤ If f is invertible then f must be bijective i.e one-one and onto.

➤ If the inverse of a function exists then it is said to be invertible. The inverse of a function if exists then it is unique.

➤ Graph of $y = f(x)$ and $y = f^{-1}(x)$ are symmetrical about $y = x$ and intersects on the line $y = x$, and also $f(x) = f^{-1}(x) = x$ when ever graph intersects.

➤ **Working Rule to find the inverse of a function**

Let $f: X \rightarrow Y$ be a bijection function.

put $f(x) = y$. Solve the equation $y = f(x)$ to obtain x in terms of y . Interchange x and y to obtain the inverse of f .

W.E.5: $f: R \rightarrow (-\infty, 1)$ given by $f(x) = 1 - 2^{-x}$.

Then $f^{-1}(x)$ is

Sol: Here $f(x)$ is both one-one and onto

Let $y = 1 - 2^{-x}$ or $2^{-x} = 1 - y$ or

$-x = \log_2(1 - y)$ or

$f^{-1}(x) = g(x) = -\log_2(1 - x)$

➤ **Properties of composite function :**

i) If $f: A \rightarrow B$, $g: B \rightarrow C$ are one-one then $g \circ f: A \rightarrow C$ is also one-one.

ii) If $f: A \rightarrow B$, $g: B \rightarrow C$ are onto then $g \circ f: A \rightarrow C$ is also onto.

iii) If $g \circ f: A \rightarrow C$ is one-one then f is one-one.

iv) If $g \circ f: A \rightarrow C$ is onto then g is onto.

v) If $f: A \rightarrow B$ is a function then

$$f \circ I_A = I_B \circ f = f.$$

vi) If $f: A \rightarrow B$ and $g: B \rightarrow C$ are bijections

then $g \circ f: A \rightarrow C$ is a bijection &

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

vii) If $f: A \rightarrow B$ is a bijection then

$$f \circ f^{-1} = I_B, f^{-1} \circ f = I_A.$$

viii) If $f: A \rightarrow B$ and $g: B \rightarrow A$ are functions such that $g \circ f = I_A$ and $f \circ g = I_B$ then $g = f^{-1}$.

ix) If $f: A \rightarrow A$ is a bijection then

$$f \circ f^{-1} = f^{-1} \circ f = I_A.$$

x) In general $f \circ g \neq g \circ f$.

xi) If $f: A \rightarrow B$, $g: B \rightarrow C$ and $h: C \rightarrow D$ are functions then $h \circ (g \circ f) = (h \circ g) \circ f$.

➤ **Even & odd functions:**

$f: A \rightarrow R, A \subseteq R$ is said to be an even function if

$$f(-x) = f(x), \forall x \in A.$$

E.g: $y = |x|$, $y = x^2$, $y = \cos x$ are some even functions.

➤ **Note:** (i) The graph of an even function is symmetric about Y-axis.

(ii) If (x, y) is point on the even function then

$(-x, y)$ is also point on even function.

➤ $f: A \rightarrow R, A \subseteq R$ is said to be an odd function if

$$f(-x) = -f(x) \quad \forall x \in A.$$

E.g: $y = x$, $y = x^3$, $y = \sin x$ are some odd functions.

➤ **Note:** (i) The graph of an odd function is symmetric about origin (symmetrical in opposite quadrants)

(ii) If (x, y) is a point on an odd function graph

then $(-x, -y)$ is also a point on the same odd function graph.

➤ **Important points of odd and even functions:**

- (i) A function which is even or odd, when even power is always even function.
- (ii) The derivative of an odd function is an even function and derivative of an even function is an odd function.
- (iii) Every function can be uniquely expressed as the sum of an even and an odd function. i.e.,

$$f(x) = \frac{1}{2}\{f(x) + f(-x)\} + \frac{1}{2}\{f(x) - f(-x)\}$$

$$= \{\text{even function}\} + \{\text{odd function}\}$$

- **Note:** (i) A function may be neither even nor odd. for example $f(x) = \sin x + x^2$ is neither even nor odd.
- (ii) $f(x) = 0$ is the function which is both even and odd.
- (iii) $f(x) = c$ is an even function
- (iv) Every even function $y = f(x)$ is not one-one $\forall x \in D_f$.

W.E.6: Identify whether the function

$f(x) = xg(x)g(-x) + \tan(\sin x)$ is odd or even.

Sol: $f(-x) = (-x)g(-x).g(x) + \tan(\sin(-x))$
 $= -(xg(x)g(-x) + \tan(\sin x)) = -f(x)$

Hence $f(x)$ is an odd function.

- **Polynomial function:** If $f: R \rightarrow R$ is defined by
 $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n$,
 $a_0 \neq 0, a_1, a_2, \dots, a_n \in R$, n is a non-negative integer is a polynomial function of degree n in x
- E.g:** (i) $x^4 - x + 2$ is a polynomial function.
(ii) $x^4 - \sqrt{x} + 2$ is not a polynomial function
- **Rational Function:** A function of the form $\frac{f(x)}{g(x)}$ where $f(x)$ and $g(x)$ are polynomial functions and $g(x) \neq 0$ is called a rational function.
- **Algebraic functions:** A function f is said to be an algebraic function if it arises due to a finite number of fundamental operations like addition, subtraction, multiplication, division and

root extraction etc, on polynomial functions.

- **Transcendental function:** The functions which are not algebraic are called transcendental functions.

E.g: i) Exponential functions
ii) Logarithmic functions
iii) Trigonometric functions
iv) inverse trigonometric functions

E.g: $f(x) = x^3 + 3x^2 - \sqrt{x^2 - x} + \sqrt[3]{x}$ is an

algebraic function where as $g(x) = x + \sin^{-1} x$.

$h(x) = e^x, \cosh^{-1} x$ are transcendental functions.

- **Algebra of real valued functions :**

➤ **Definition :** Let $f: A \rightarrow B$ is a function and

i) if $A \subseteq R$ then f is called a real variable function.

ii) if $B \subseteq R$ then f is called a real valued function.

iii) if $A \subseteq R, B \subseteq R$ then f is called a real function.

- **Properties:** If f and g are real valued functions with domain A and B respectively, then both f and g are defined on $A \cap B$ when $A \cap B \neq \phi$.

➤ Let $f: A \rightarrow R$ & $g: B \rightarrow R$ then

(i) $(f \pm g)(x) = f(x) \pm g(x), \forall x \in A \cap B$

(ii) $(fg)(x) = f(x).g(x), \forall x \in A \cap B$

(iii) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \forall x \in A \cap B \text{ \& } g(x) \neq 0$

(iv) $(f \pm k)(x) = f(x) \pm k, k \in R$

(v) $(kf)(x) = kf(x), k \in R$

(vi) $f^n(x) = \{f(x)\}^n, n > 0$

(vii) $|f|(x) = |f(x)|, x \in A$

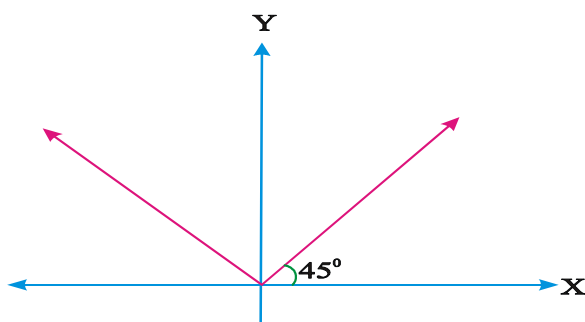
- **Modulus function(Absolute value function):**

The absolute value or numerical value or the modulus of real number x denoted by $|x|$ is defined as

$$|x| = \begin{cases} -x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ x & \text{if } x > 0 \end{cases}$$

Thus we have $|x| \geq 0$ and $|-x| = |x|$

➤ The graph of $f(x) = |x|$ is



➤ The domain of $|x|$ is $\mathbb{R}$ and range is $[0, \infty)$

➤ **Properties of modulus function:**

(i) $x^2 = |x|^2 = |x^2|$

(ii) $|xy| = |x||y|$

(iii) $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$, provided $y \neq 0$

(iv) $|x| \leq a \Rightarrow -a \leq x \leq a; (a \geq 0)$

(v) $|x| \geq a \Rightarrow x \leq -a$ and $x \geq a; (a \geq 0)$

(vi) $|x \pm y| \leq |x| + |y|$ and if $xy \geq 0$ then $|x + y| = |x| + |y|$

(vii) $|x - y| = |x| - |y| \Rightarrow x \geq 0, y \geq 0, |x| \geq |y|$

(or)

$$x \leq 0, y \leq 0, |x| \geq |y|$$

(viii) $|x \pm y| \leq |x| + |y|$

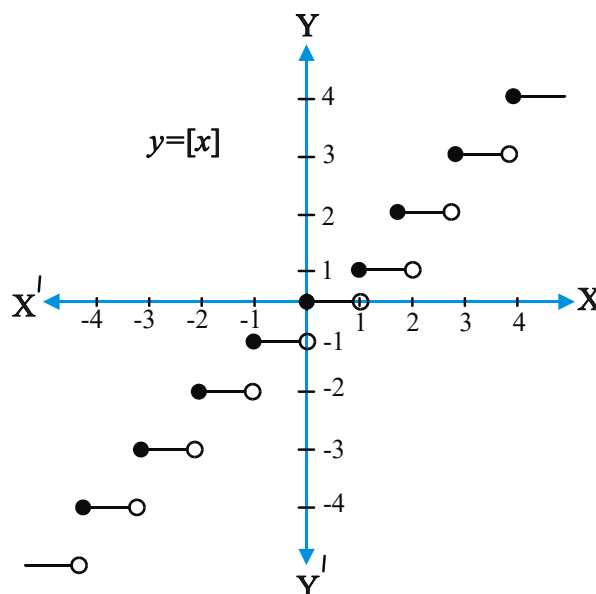
(ix) $|x \pm y| \geq ||x| - |y||$

➤ **Step function (Greatest integer function) (or) floor function :**

The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = [x]$ is called the greatest integer function, where

$[x]$ equal to integral part of x or greatest integer less than or equal to x . (or) If x is any real number then there exist integers n and $n+1$ such that $n \leq x < n+1$. Then the integral part of x is defined

as n . It is denoted by $[x]$.



➤ From the definition of $[x]$, we have

(i) $[x] = -1$, for $-1 \leq x < 0$

(ii) $[x] = 0$, for $0 \leq x < 1$

(iii) $[x] = 1$, for $1 \leq x < 2$

(iv) $[x] = 2$, for $2 \leq x < 3$

➤ The domain of $[x]$ is $\mathbb{R}$ and range is $\mathbb{Z}$

➤ **Properties of greatest integer function:**

(i) If $f(x) = [x + n] = [x] + n$, where $n \in \mathbb{I}$ and $[.]$ denotes the greatest integer function, then

$$f(x) = n + [x]$$

(ii) $x - 1 < [x] \leq x$

(iii) $[x] + [-x] = \begin{cases} 0 & \text{if } x \in \mathbb{I} \\ -1 & \text{if } x \notin \mathbb{I} \end{cases}$

(iv) $[x] - [-x] = 2n - 1$, if $n - 1 < x < n$, $n \in \mathbb{I}$

(v) $[x] - [-x] = 2n + 1$, if $x = n + \{x\}$, $n \in \mathbb{I}$

(vi) $[x + y] \geq [x] + [y]$.

(vii) $[x] \geq k \Rightarrow x \geq k$, where $k \in \mathbb{Z}$

(viii) $[x] \leq k \Rightarrow x < k + 1$, where $k \in \mathbb{Z}$

(ix) $[x] > k \Rightarrow x \geq k + 1$, where $k \in \mathbb{Z}$

$$(x) [x] < k \Rightarrow x < k, \text{ where } k \in \mathbb{Z}$$

$$(xi) [x] \leq x < [x] + 1$$

$$(xii) [x + y] = [x] + [y] \text{ if either } x \text{ (or) } y \text{ (or) both are integers}$$

$$(xiii) [[x]] = [x]$$

$$(xiv) \left[\frac{x}{n} \right] + \left[\frac{x+1}{n} \right] + \left[\frac{x+2}{n} \right] + \dots + \left[\frac{x+n-1}{n} \right] = [x], n \in \mathbb{N}$$

$$(xv) [x] + \left[x + \frac{1}{n} \right] + \left[x + \frac{2}{n} \right] + \dots + \left[x + \frac{n-1}{n} \right] = [nx]$$

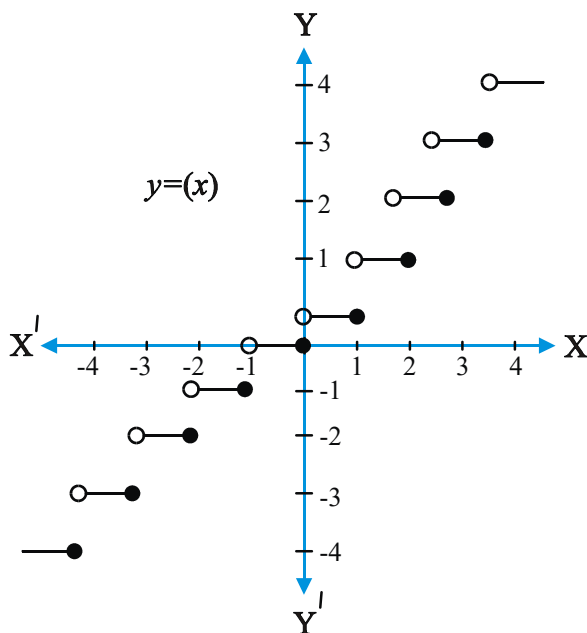
➤ **Least Integer Function:**

If $x \in \mathbb{R}$ then the least integer greater than or equal to x is denoted by (x) is called least integer function or ceiling function.

Eg : (i) $(3.2) = 4,$

(ii) $(-2.3) = -2,$

(iii) $(5) = 5$



$$i) (x) = \begin{cases} -1, & -2 < x \leq -1 \\ 0, & -1 < x \leq 0 \\ 1, & 0 < x \leq 1 \\ 2, & 1 < x \leq 2 \end{cases}$$

ii) Domain of (x) is $\mathbb{R}$ and Range is $\mathbb{Z}$.

Properties of L.I.F

i) $((x)) = x$

$$ii) (x) = \begin{cases} [x] = x; & x \in \mathbb{I} \\ [x] + 1; & x \notin \mathbb{I} \end{cases}$$

$$iii) (x) + (-x) = \begin{cases} 0; & x \in \mathbb{I} \\ 1; & x \notin \mathbb{I} \end{cases}$$

$$iv) (x) - (-x) = \begin{cases} 2x; & x \in \mathbb{I} \\ 2(x) - 1; & x \notin \mathbb{I} \end{cases}$$

$$v) (x) = \left(\frac{x}{2} \right) + \left(\frac{x-1}{2} \right)$$

$$vi) (x) + \left(x + \frac{1}{n} \right) + \left(x + \frac{2}{n} \right) + \dots + \left(x + \frac{n-1}{n} \right) = (nx) + n - 1, n \in \mathbb{N}$$

$$vii) \left(\frac{n+1}{2} \right) + \left(\frac{n+2}{4} \right) + \left(\frac{n+4}{8} \right) + \left(\frac{n+8}{16} \right) + \dots + \left(\frac{n+2^{n-1}}{2^n} \right) = 2n - 1, n \in \mathbb{N}$$

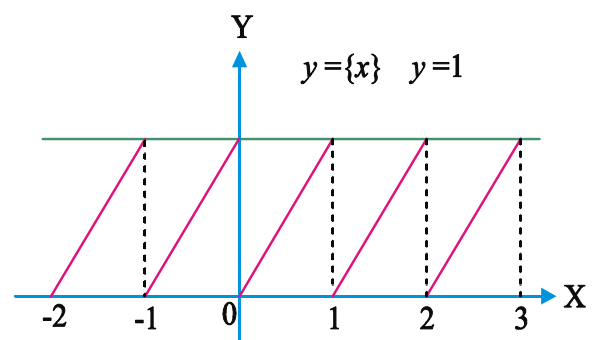
➤ **Fractional Part of x:**

For $x \in \mathbb{R}$, the function $f(x) = x - [x]$ is called fractional part of x , it is denoted by $\{x\}$ or $\{x\}$.

and is defined by $\{x\} = f$ if $x = n + f$

where $n \in \mathbb{I}$ and $0 \leq f < 1$

Eg : $\{2.7\} = 0.7, \{-3.6\} = 0.4, \{3\} = 0$



➤ Domain of $\{x\}$ is $\mathbb{R}$ and Range of $\{x\}$ is $[0, 1)$

➤ The function $\{x\}$ is neither even nor odd

$$\{x\} = \begin{cases} x+2, & -2 \leq x < -1 \\ x+1, & -1 \leq x < 0 \\ x, & 0 \leq x < 1 \\ x-1, & 1 \leq x < 2 \\ x-2, & 2 \leq x < 3 \end{cases}$$

Properties of fractional part function

i) $\{x \pm n\} = \{x\}; n \in I$

ii) $\{x\} = 0; x \in I$

iii) $\{x\} + \{-x\} = \begin{cases} 0; & x \in I \\ 1; & x \notin I \end{cases}$

iv) $\{x\} - \{-x\} = \begin{cases} 0; & x \in I \\ 2\{x\} - 1; & x \notin I \end{cases}$

v) $\{[x]\} = 0$

vi) If $f(x) = \frac{1}{\{x\}}$ then $D_f = R - I$

$$R_f = (1, \infty)$$

Eg: Solve the equation $4\{x\} = x + [x]$

Sol. $4\{x\} = [x] + \{x\} + [x]$

$$\Rightarrow \{x\} = \frac{2}{3}[x] \Rightarrow 0 \leq \frac{2}{3}[x] < 1$$

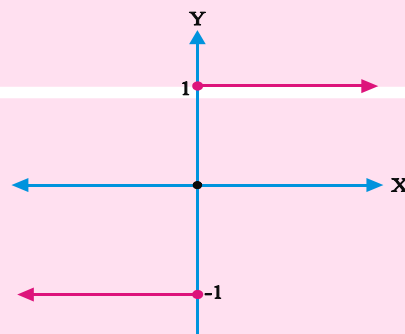
$$0 \leq [x] < \frac{3}{2}$$

$$\Rightarrow \{x\} = 0, \frac{2}{3} \Rightarrow x = 0, \frac{5}{3}$$

➤ **Signum Function:** The signum function or signature function is defined as

$$\text{Sgn}(x) = \begin{cases} \frac{|x|}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

$$\text{i.e. } \text{Sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases}$$



➤ The domain of $\text{Sgn}(x)$ is R and its

Range is $\{-1, 0, 1\}$

Properties of signum function

i) $\text{sgn}(Kx) = \text{sgn}(x); K \in N$

ii) $x \cdot \text{sgn}(x) = |x|$

iii) $|x| \cdot \text{sgn}(x) = x$

iv) $x \cdot \text{sgn}(x) \cdot \text{sgn}(x) = x$

W.E-7: Find the domain and range of

$$f(x) = \text{sgn}(x^2 + 1)$$

Sol:

$$f(x) = \text{sgn}(x^2 + 1) = \begin{cases} 1; & x^2 + 1 > 0 \\ -1; & x^2 + 1 < 0 \text{ (Not possible)} \\ 0; & x^2 + 1 = 0 \text{ (Not possible)} \end{cases}$$

$$\therefore f(x) = 1, x \in R \Rightarrow D_f = R; R_f = 1$$

➤ **Linear function:** The function f defined by $f(x) = mx + c, x \in R$, is called linear function, where m and c are constants.

➤ **Explicit function:** A function $y = f(x)$ is said to be an explicit function of x if the dependent variable y can be expressed in terms of the independent variable x .

E.g: i) $y = x - \cos x$, ii) $y = x + \log_e x - 2x^3$.

- **Implicit function:** A function $y = f(x)$ is said to be an implicit function of x if the dependent variable y can not be expressed in terms of the independent variable x only.

E.g: i) $3x^2 - 4x^2y^2 + 2xy = 7$ ii) $xy = \cos(x+y)$

- **Periodic function:** A function $f(x)$ is said to be periodic function if there exists a positive real number T , such that $f(x+T) = f(x)$, $\forall x \in D_f$, where the least positive real number T is called its fundamental period.

W.E.8: The period (if periodic) of the function

$f(x) = x - [x - b]$, $b \in R$ ($[.]$ denotes the greatest integer function) is

Sol: $f(x) = x - [x - b]$

$$x - b - [x - b] + b = b + \{x - b\}$$

So, $f(x)$ has period 1. ($\because$ Period $\{x\}$ is 1)

- **Extension of a function:**

Let a function $f(x)$ defined on the interval $[0, a]$, it can be extended on $[-a, a]$ so that $f(x)$ is either even or odd function on $[-a, a]$

- (i) **Even Extension**

As $f(x)$ is defined on $[0, a]$

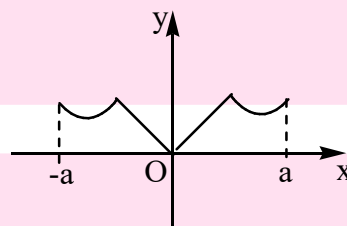
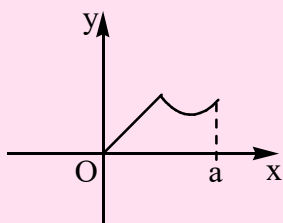
$f(-x)$ is defined on $[-a, 0]$, then even

extension i.e., $F_e(x)$ is defined x^6

$$F_e(x) = f(x); x \in [0, a]$$

$$= f(-x); x \in [-a, 0]$$

Graphically If the graph of $y = f(x)$, $x \in [0, a]$ is as follows then even functions is given as

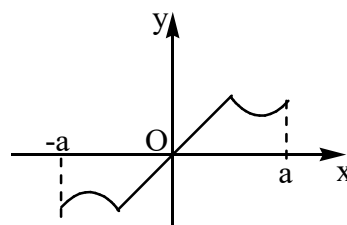


i.e., Even extension function of $y = f(x)$ is obtained as a function which is symmetrical about y-axis.

- (ii) **Odd Extension :** As $f(x)$ is defined on $[0, a]$, then odd extension i.e., $F_o(x)$ is given as

$$F_o(x) = f(x); x \in [0, a]$$

$$= -f(-x); x \in [-a, 0]$$



i.e., odd extension is obtained as a function which is symmetrical about the origin.

W.E-9 : Let $f(x) = e^x + \sin x$ be defined on $[-4, 0]$.

Find the odd and Even Extensions of $f(x)$ is $[-4, 4]$

Sol. Odd extension = $F_o(x) =$

$$\begin{cases} f(x); & x \in [-4, 0] \\ -f(-x); & x \in [0, 4] \end{cases} = \begin{cases} e^x + \sin x; & x \in [-4, 0] \\ -e^{-x} + \sin x; & x \in [0, 4] \end{cases}$$

Even extension = $F_e(x) =$

$$\begin{cases} f(x); & x \in [-4, 0] \\ f(-x); & x \in [0, 4] \end{cases} = \begin{cases} e^x + \sin x; & x \in [-4, 0] \\ e^{-x} - \sin x; & x \in [0, 4] \end{cases}$$

W.E-10: Let $f(x) = \begin{cases} x^3 + x^2; & 0 \leq x \leq 2 \\ -x + 2; & 2 \leq x < 4 \end{cases}$

Even extension =

$$F_e(x) = \begin{cases} f(x); & x \in [0, 4] \\ f(-x); & x \in [-4, 0] \end{cases}$$

$$= \begin{cases} x^3 + x^2; & 0 \leq x \leq 2 \\ x + 2; & 2 < x \leq 4 \\ -x^3 + x^2; & -2 \leq x \leq 0 \\ -x + 2; & -4 \leq x \leq -2 \end{cases}$$

Odd Extension =

$$F_e(x) = \begin{cases} f(x); & x \in [0, 4] \\ f(-x); & x \in [-4, 0] \end{cases}$$

$$= \begin{cases} x^3 + x^2; & 0 \leq x \leq 2 \\ x + 2; & 2 < x \leq 4 \\ x^3 - x^2; & -2 \leq x \leq 0 \\ x - 2; & -4 \leq x \leq -2 \end{cases}$$

W.E.11: The domain of $f(x) = \sqrt{\sin^{-1}(\log_2 x)}$ is

Sol: $\sin^{-1}(\log_2 x) \geq 0$ and $-1 \leq \log_2 x \leq 1$ and $x > 0$
 $\log_2 x \geq 0$ & $\log_2 x \leq 1$ & $x > 0$.
 $0 \leq \log_2 x \leq 1, x > 0$
 $2^0 \leq x \leq 2^1 \quad \therefore x \in [1, 2]$

➤ **Range:** The set of values of 'x' for which $f(x)$ is defined is called the domain of $f(x)$ and is denoted by D_f . The set of images of the elements of the domain (out comes) is called the range of $f(x)$ and is denoted by R_f .

Study of range is very important. There are no fixed methods to find range. Some of the methods to find the range are given as follows.

Type (1) Usage of $AM \geq GM \geq HM$ on the given positive quantities

Eg: Find the range of $x + \frac{1}{x}$

Sol. Let $x > 0, AM \geq GM$

$$x + \frac{1}{x} \geq 2 \Rightarrow \sqrt{x \cdot \frac{1}{x}} \Rightarrow x + \frac{1}{x} \geq 2$$

Let $x < 0$, Let $x = -y$ where $y > 0$

$$y + \frac{1}{y} \geq 2 \Rightarrow -x - \frac{1}{x} \geq 2$$

$$\Rightarrow x + \frac{1}{x} \leq -2$$

$$\therefore \text{Range} = (-\infty, -2] \cup [2, \infty)$$

Type (2) The range of $y = ax^2 + bx + c$ is

$$\left[\frac{4ac - b^2}{4a}, \infty \right) \text{ (or) } \left(-\infty, \frac{4ac - b^2}{4a} \right]$$

according as $a > 0$ or $a < 0$

Type (3) To find the range of a function whose domain is $(-\infty, \infty)$, we proceed as follows. Let $y = f(x)$

Now express 'x' in terms of 'y' i.e., $x = g(y)$ (if possible)

Now find 'y' for which 'x' is defined.

Type (4) To find the range f of $f(x)$ in $[a, b]$, find

$f'(x)$ and put $f'(x) = 0 \Rightarrow x = \alpha, \beta, \gamma, \dots$
 accept these values if they lie in $[a, b]$ find these images and also find $f(a), f(b)$. The least, greatest values thus obtained are called Max, Min. values of $f(x)$ in $[a, b]$

To find range of $f(x)$ in (a, b)

follow the above procedure is $f'(x) = 0 \Rightarrow x = \alpha, \beta, \gamma, \dots$

If m, M are the least, greatest values.

If $\lim_{x \rightarrow a^+} (f(x))$ and $\lim_{x \rightarrow b^-} (f(x))$ both are greater than 'm' then 'm' is least value. Similarly,

if $\lim_{x \rightarrow a^+} (f(x))$ and $\lim_{x \rightarrow b^-} (f(x))$ both are less than 'M', then M is the greatest value. Otherwise we cannot decide the least, greatest values.

Type (5) To find the range of $f(x) = \frac{P(x)}{Q(x)}$, where

degree is '2' then $yQ(x) = P(x) \Rightarrow$ we have a quad. equation in 'x'

$\Delta \geq 0$ ($\because$ 'x' is real) $\Rightarrow$ range will come

W.E-12: Find the range of $y = \frac{x^2 + 2x + 3}{x}$

Sol. $x^2 + (2 - y)x + 3 = 0$

$$\Rightarrow \Delta \geq 0 \Rightarrow (2 - y)^2 - 4(3) \geq 0$$

$$\Rightarrow 4 + y^2 - 4y - 12 \geq 0 \Rightarrow y^2 - 4y - 8 \geq 0$$

$$\Rightarrow y \in (-\infty, 2 - 2\sqrt{3}] \cup [2 + 2\sqrt{3}, \infty)$$

Type (6) If various functions of different natures are present in the given problem, then find the common domain and have the range.

Type (7) If $f(x)$ is strictly increasing function in $[a, b]$

then its range $= [f(a), f(b)]$

If $f(x)$ is strictly decreasing in $[a, b]$ then its

range $= [f(b), f(a)]$

W.E.13: The range of $f(x) = \sin^{-1} x + \tan^{-1} x$ is

Sol: Domain of $f(x)$ is $[-1, 1]$

$$\sin^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \tan^{-1} x \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$$

$$-\frac{\pi}{2} - \frac{\pi}{4} \leq \sin^{-1} x + \tan^{-1} x \leq \frac{\pi}{2} + \frac{\pi}{4}$$

$$-\frac{3\pi}{4} \leq f(x) \leq \frac{3\pi}{4}$$

W.E.14: The range of $f: A \rightarrow B$ where

$A = \{1, 2, 3, 4\}$ and $f(x)$ is defined as

$$f(x) = x^2 + x + 1 \text{ is}$$

Sol. $f(1) = 1 + 1 + 1 = 3$

Similarly $f(2) = 7, f(3) = 13, f(4) = 21$

$$\text{Range} = \{3, 7, 13, 21\}$$

ii) If the domain of $y = f(x)$ is $\mathbb{R}$ (i.e. the set of real numbers) or $\mathbb{R}$ -{some finite points} or an infinite interval, express x in terms of y. From this,

(a) find y for x to be defined or real, or

(b) If a quadratic equation is formed in terms of x then apply the condition of real roots. ($\Delta \geq 0$)

W.E.15: The Range of $f(x) = \frac{x^2 + x - 3}{x - 1}$ is

Sol: $f(x) = \frac{x^2 + x - 3}{x - 1} = y \Rightarrow xy - y = x^2 + x - 3$

$$x^2 + x(1 - y) + y - 3 = 0$$

Quadratic eq. in terms of x and $x \in \mathbb{R}$

$$\therefore b^2 - 4ac \geq 0, (1 - y)^2 - 4(y - 3)(1) \geq 0$$

$$(y - 3)^2 + 4 \geq 0 \quad y \in \mathbb{R} \therefore \text{range is } \mathbb{R}.$$

W.E.16: The range of the following functions defined over $[0, 3]$

$$f(x) = \begin{cases} x; & 0 \leq x \leq 1 \\ x^2 + 1; & 1 < x \leq 2 \\ x^3; & 2 < x \leq 3 \end{cases} \text{ is}$$

Sol: Obviously the domain of the function is $[0, 3]$.

Its range is $[0, 1]$ for $0 \leq x \leq 1$;

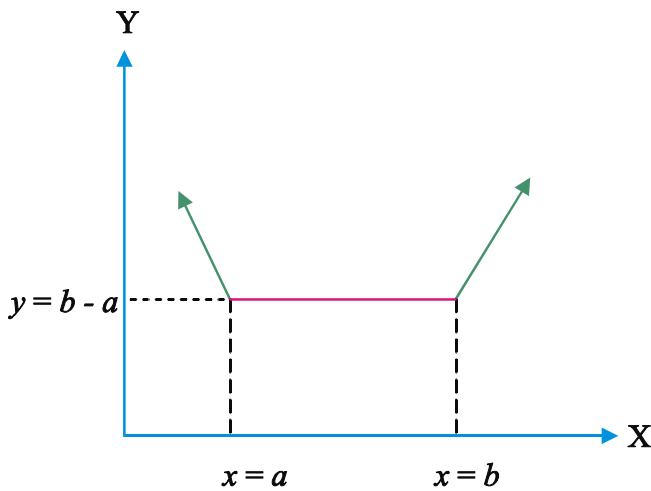
$(2, 5]$ for $1 < x \leq 2$ and $(8, 27]$ for $2 < x \leq 3$.

$$\therefore \text{Range} = [0, 1] \cup (2, 5] \cup (8, 27].$$

iii) If the graph of the function is known, then the range of the function can be obtained easily from the graph.

W.E.17: The range of $|x - a| + |x - b|$ ($b > a$) is

Sol:



from figure we conclude range of $f(x)$ is $[b - a, \infty)$.

iv) By using $AM \geq GM \geq HM$ concept.

W.E.18: The range of the function

$f(x) = 6^x + 3^x + 6^{-x} + 3^{-x} + 2$ is

Sol: $6^x + 6^{-x} \geq 2\sqrt{6^x \cdot 6^{-x}} = 2$

$3^x + 3^{-x} \geq 2\sqrt{3^x \cdot 3^{-x}} = 2$

$\therefore f(x) = 6^x + 3^x + 6^{-x} + 3^{-x} + 2 \geq 2 + 2 + 2,$

$f(x) \geq 6$

$\therefore$ range of $f(x)$ is $[6, \infty)$

v) Range of $f(x) = ax^2 + bx + c$ is $\left[-\frac{D}{4a}, \infty\right)$

if $a > 0$ and $\left(-\infty, -\frac{D}{4a}\right]$ if $a < 0$,

where $D = b^2 - 4ac$.

vi) If $a^2 + b^2 + c^2 = k$, then range of

$ab + bc + ca$ is $\left[-\frac{k}{2}, k\right]$

➤ **Domain and range of some standard functions:**

S.No	Function	Domain	Range
1.	$\sqrt{a^2 - x^2}$	$[-a, a]$	$[0, a]$
2.	$\frac{1}{\sqrt{a^2 - x^2}}$	$(-a, a)$	$\left[\frac{1}{a}, \infty\right)$

3.	$\sqrt{x^2 - a^2}$	$R - (-a, a)$	$[0, \infty)$
4.	$\frac{1}{\sqrt{x^2 - a^2}}$	$R - [-a, a]$	$(0, \infty)$
5.	$a^x, (a > 0, a \neq 1)$	R	$(0, \infty)$
6.	e^x	R	$(0, \infty)$
7.	$\log_a x$ ($a > 0, a \neq 1$)	$(0, \infty)$	R
8.	$ x $	R	$[0, \infty)$
9.	$[x]$	R	Z
10.	$\{x\} = x - [x]$	R	$[0, 1)$
11.	$\sqrt{x}$	$[0, \infty)$	$[0, \infty)$
12.	$\sin x$	R	$[-1, 1]$
13.	$\cos x$	R	$[-1, 1]$
14.	$\tan x$	$R - \left\{(2n+1)\frac{\pi}{2} : n \in Z\right\}$	R
15.	$\operatorname{cosec} x$	$R - \{n\pi : n \in Z\}$	$R - (-1, 1)$
16.	$\sec x$	$R - \left\{(2n+1)\frac{\pi}{2} : n \in Z\right\}$	$R - (-1, 1)$
17.	$\cot x$	$R - \{n\pi : n \in Z\}$	R
18.	$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
19.	$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
20.	$\tan^{-1} x$	R	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
21.	$\operatorname{Cosec}^{-1} x$	$R - (-1, 1)$	$\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$
22.	$\sec^{-1} x$	$R - (-1, 1)$	$\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$
23.	$\cot^{-1} x$	R	$(0, \pi)$
24.	$a \cos x + b \sin x + c$	R	$\left[c - \sqrt{a^2 + b^2}, c + \sqrt{a^2 + b^2}\right]$
25.	$\sinh x$	R	R
26.	$\cosh x$	R	$[1, \infty)$
27.	$\tanh x$	R	$(-1, 1)$
28.	$\operatorname{cosech} x$	$R - \{0\}$	$R - \{0\}$
29.	$\operatorname{sech} x$	R	$(0, 1]$
30.	$\coth x$	$R - \{0\}$	$R - [-1, 1]$
31.	$\sinh^{-1} x$	R	R

32. $\cosh^{-1}x$ $[1, \infty)$ $[0, \infty)$
 33. $\tanh^{-1}x$ $(-1, 1)$ $\mathbb{R}$
 34. $\operatorname{cosech}^{-1}x$ $\mathbb{R} - \{0\}$ $\mathbb{R} - \{0\}$
 35. $\operatorname{sech}^{-1}x$ $(0, 1]$ $[0, \infty)$
 36. $\operatorname{coth}^{-1}x$ $\mathbb{R} - [-1, 1]$ $\mathbb{R} - \{0\}$

➤ **Functional equations:** If $f(x)$ is a function such that

i) $f(x+y) = f(x)f(y)$ then $f(x) = k^x$

$(k \in \mathbb{R}^+)$

ii) $f(x+y) = f(x) + f(y)$ then $f(x) = kx$

iii) $f(xy) = f(x) + f(y)$ then

$f(x) = k \log_a x$ ($a \neq 1, a > 0$)

iv) If $f(x)$ is a polynomial function such that

$f(x) + f\left(\frac{1}{x}\right) = f(x)f\left(\frac{1}{x}\right)$ then $f(x) = 1 \pm x^n$.

v) If $f(x)$ is a function such that

$f(x+y) + f(x-y) = 2f(x)f(y)$

then $f(x) = \frac{k^x + k^{-x}}{2}$ or $f(x) = \cos x$

➤ **Some more points to observe :**

- Any function, which is entirely increasing or decreasing in its whole domain, is said to be one-one function
- If any line parallel to the X-axis cuts the graph of the function at one point, then the function is one-one
- Any function which is neither increasing nor decreasing in whole domain, then $f(x)$ is many-one. (OR) any continuous function $f(x)$ which has at least one local maxima or local minima, is many-one
- Every odd continuous function passes through the origin and it is symmetrical in opp. quadrants
- $\sin^n x, \cos^n x, \sec^n x, \operatorname{cosec}^n x$ periodic functions with period 2π (or) π according as 'n' is odd or even
- $\tan^n x, \cot^n x$ are periodic functions with period π , if 'n' is even or odd
- $|\sin x|, |\cos x|, |\tan x|, |\cot x|, |\sec x|, |\operatorname{cosec} x|$ are periodic functions with the period π

8. Algebraic functions i.e.,

$x^2, 2x^2 + 5x + 4, \sqrt{x}, \dots$ are not periodic

9. Every constant function is always periodic with no fundamental period

10. A function can have infinite periods, but among them the least positive value is called the fundamental period

11. If $f(x)$ is the periodic function with the period T, then the function $f(ax+b)$ is periodic with the

period $\frac{T}{|a|}$

12. If $f(x)$ is periodic with T, then

$\frac{1}{f(x)}$ and $\sqrt{f(x)}$ are also periodic with the same period T

13. If $f(x)$ is periodic with period T then

$cf(x), f(x \pm c), f(x) \pm c$ are also periodic with the period T where 'c' is any constant

14. Inverse of a periodic function does not exist. But here by restricting the domain, we can have inverse

15. Strictly increasing and strictly decreasing functions are non-periodic

16. If the given problem is a combination of two or more functions, then find their periods separately and take their L.C.M (if L.C.M is possible). Here L.C.M is the period. If L.C.M is not possible, then period does not exist

17. L.C.M of

$\frac{a}{b}, \frac{c}{d}, \frac{e}{f}, \dots = \frac{\text{L.C.M of } a, c, e, \dots}{\text{H.C.F of } b, d, f, \dots}$

18. L.C.M of a rational number with irrational number is not possible

19. If $h(x) = f_1(x) + f_2(x)$ where T_1 and T_2 are the periods of $f_1(x)$ and $f_2(x)$

Now period of $h(x) = \text{L.C.M of } T_1, T_2$

$= \frac{1}{2}(\text{L.C.M of } T_1, T_2)$ if $f_1(x)$ and $f_2(x)$

are even and pair wise complementary functions.

20. The graphs of $f(x)$ and its inverse $f^{-1}(x)$ are symmetrical about the line $y=x$

21. Every even function is many-one

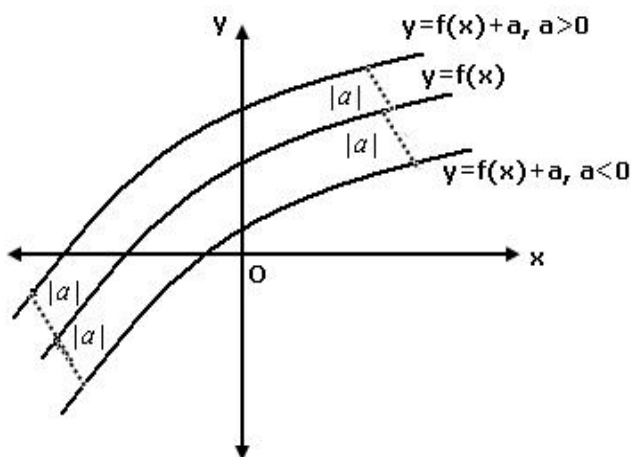
➤ **GRAPHICAL TRANSFORMATIONS :**

With the help of the given basic function (curve) $y = f(x)$, drawing the other graphs by allowing some changes in x (independent variable) and in y (dependent variable) is called transformation of graphs. Now let us observe some important transformations of graphs.

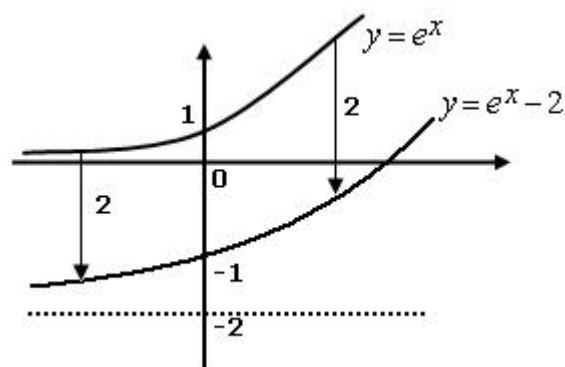
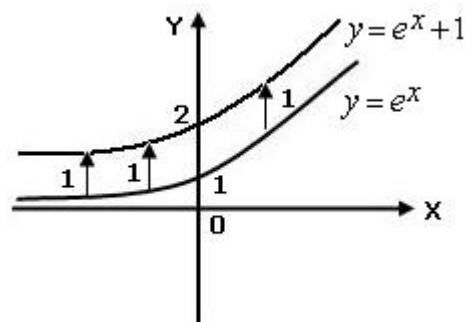
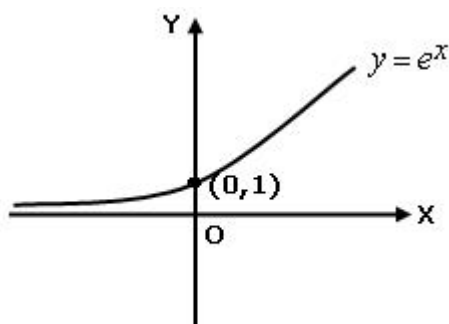
1. **Drawing the graph of $y = f(x) + a, (a \in R)$**

from the known graph of $y = f(x)$.

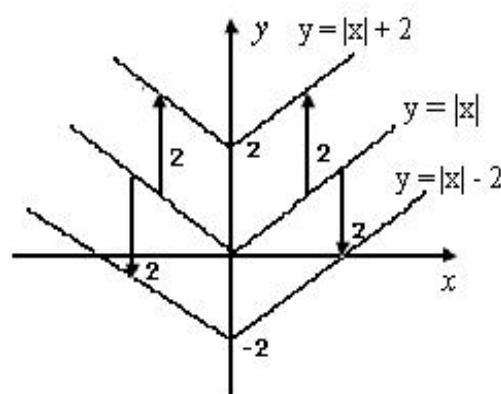
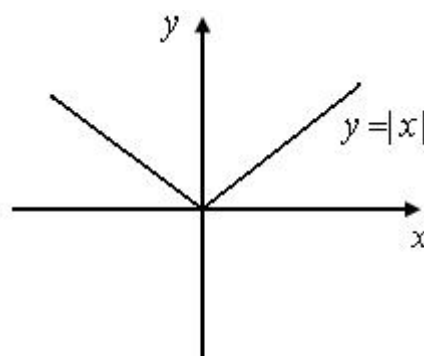
If $a > 0$ then shift the graph of $f(x)$ in upward direction through a distance 'a' (or) shift the X-axis downward through a distance 'a' and if $a < 0$ then shift the graph of $f(x)$ in downward direction through a distance 'a' or shift the X-axis upward through a distance 'a' units.



W.E-19: Draw the graph of $y = e^x + 1$, $y = e^x - 2$ with the help of $y = e^x$

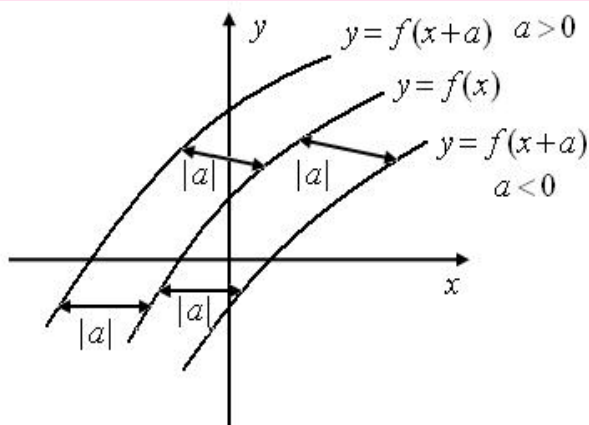


W.E-20 : plot $y = |x| + 2$ and $y = |x| - 2$ with the help of $y = |x|$

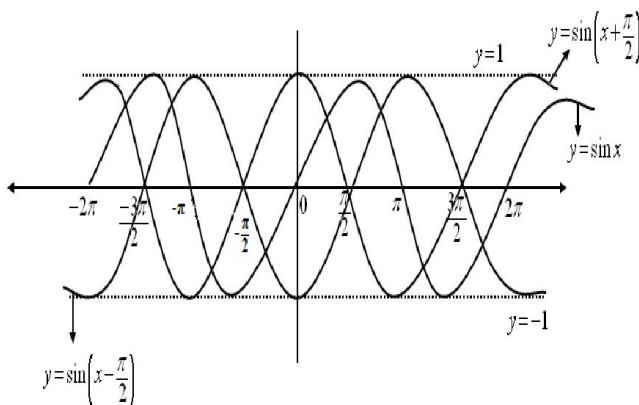


2. Drawing the graph of $y = f(x+a)$ ($a \in R$) with the help of the graph $y = f(x)$

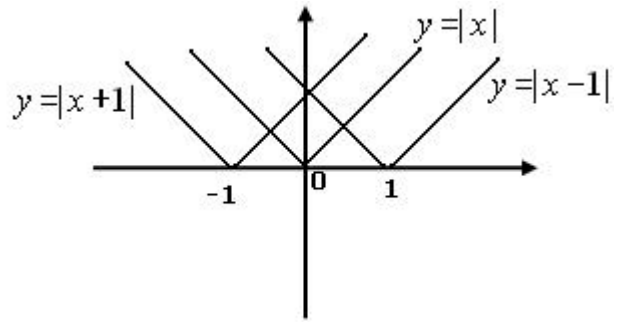
If $a > 0$, then shift the graph of $f(x)$ in the left hand direction through a distance 'a' (or) shift the y-axis to the right hand direction by distance 'a'.
Similarly if $a < 0$ then shift the graph of $f(x)$ in the right hand direction (or) shift the y-axis to the left by the distance 'a' units.



W.E-21 : Plot the graph of $y = \sin\left(x - \frac{\pi}{2}\right)$ and $y = \sin\left(x + \frac{\pi}{2}\right)$ with the help of $y = \sin x$

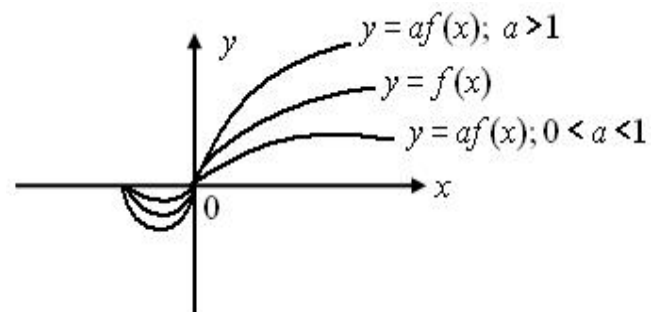


W.E-22 : Draw the graph of $y = |x - 1|$ and $y = |x + 1|$ with the help of $y = |x|$

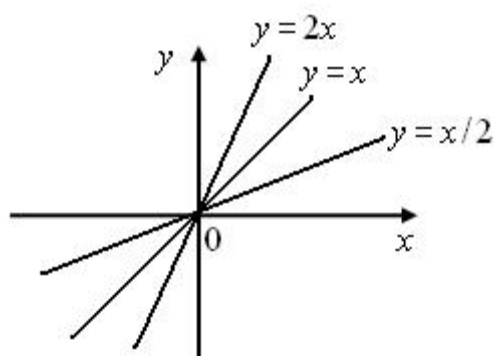


3. Drawing the graph of $y = af(x)$ from the known graph $y = f(x)$

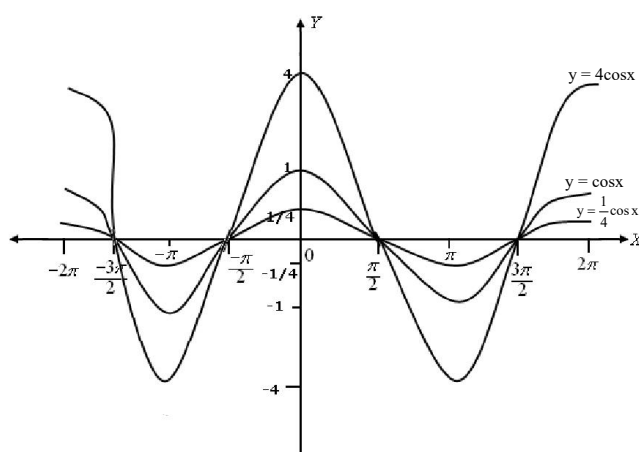
The graph of the function $y = af(x)$ ($a > 0$) is obtained from the graph of $y = f(x)$ by "stretching" vertically 'a' time ($a > 1$) and compressing vertically $\frac{1}{a}$ times ($0 < a < 1$) along the y-axis.



W.E-23 : Plot $y = 2x$ and $y = \frac{x}{2}$ with the help of $y = x$
i.e., $y = 2x$ is the stretch of $f(x)$ two times along the y-axis and $y = \frac{1}{2}x$ is the shrink of $f(x)$ two times along the y-axis.

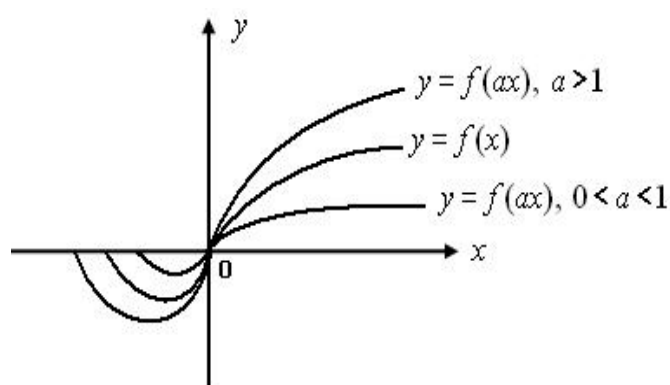


W.E- 24 : Draw the graph of $y = 4\cos x$ and $y = \frac{1}{4}\cos x$ with the help of $y = \cos x$

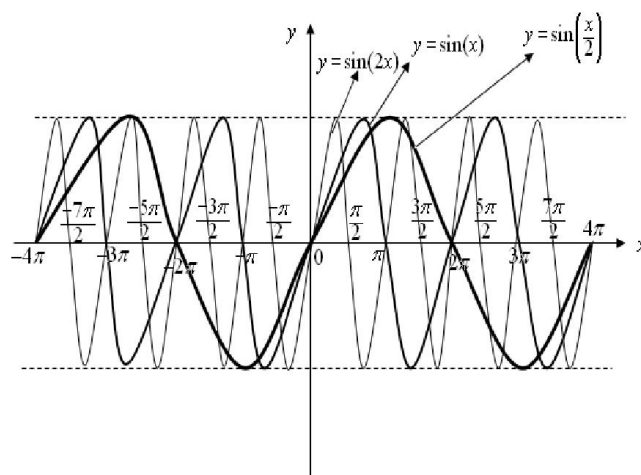


4. Drawing the graph of $y = f(ax)$ from the graph of $y = f(x)$

The graph of $y = f(ax)$ ($a > 0$) is obtained from $y = f(x)$ by stretching horizontally (along x-axis) from the y-axis $\frac{1}{a}$ times ($0 < a < 1$) and compressing horizontally from the y-axis 'a' times ($a > 1$)

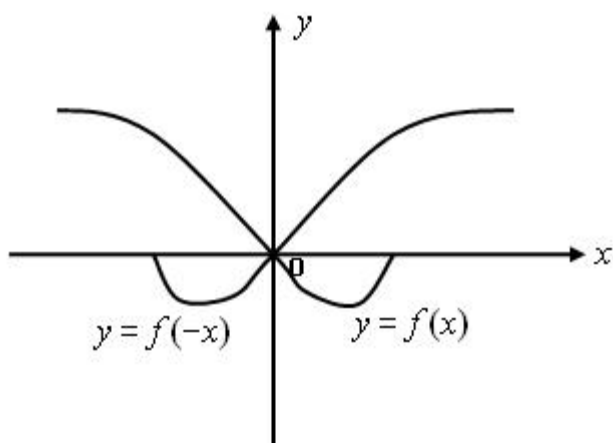


W.E-25: Draw the graph of $y = \sin(2x)$ and $y = \sin\left(\frac{x}{2}\right)$ from the graph of $y = \sin x$.

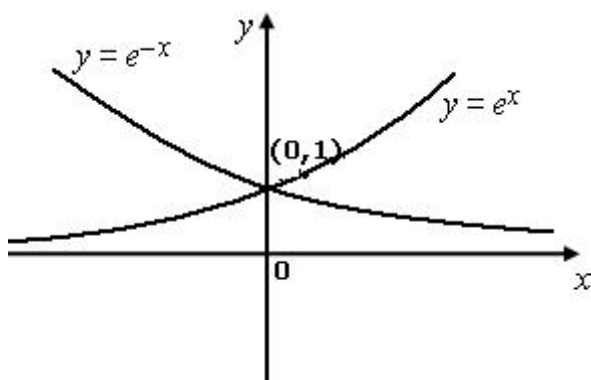


5. Drawing the graph of $y = f(-x)$ from the known graph of $y = f(x)$

The graph of $y = f(-x)$ is obtained from the graph of the function $y = f(x)$, $y = f(-x)$ is symmetrical to that of the function $y = f(x)$ about the y-axis. i.e., take reflection of $f(x)$ considering y-axis as plane mirror i.e., turn the graph of $f(x)$ by 180° about y-axis.



W.E-26: Draw the graph of $y = e^{-x}$ with the help of $y = e^x$

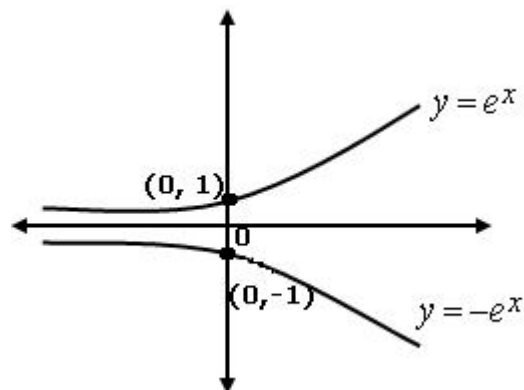


i.e., $y = e^{-x}$ is the image of $y = e^x$ w.r.t y-axis.

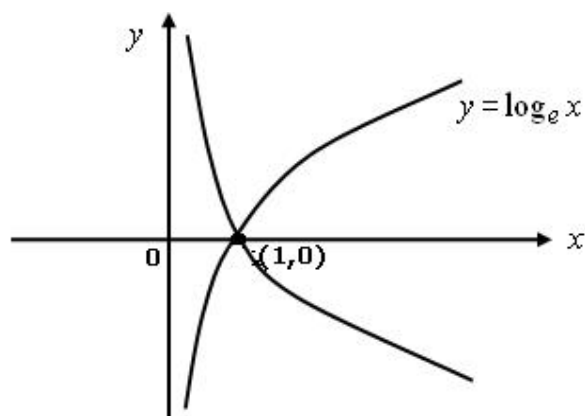
6. Drawing the graph of $y = -f(x)$ from the known graph of $y = f(x)$

The graph of $y = -f(x)$ is the image of $y = f(x)$ in the X-axis as plane mirror (OR) to get the graph of $y = -f(x)$, turn the graph of $f(x)$ by 180° about the X-axis.

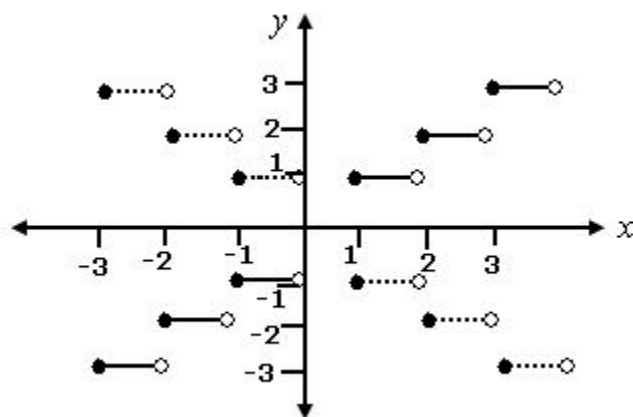
Ex. 1) Plot the curve $y = -e^x$ with the help of $y = e^x$



W.E-27: Plot the curve $y = -\log_e x$ with the help of $y = \log_e x$



W.E-28: Draw the graph of $y = -[x]$ (G.I.F) with the help of $y = [x]$ (G.I.F)

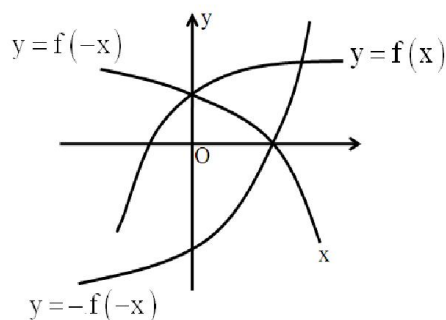


7. Drawing the graph of $y = -f(-x)$ from the

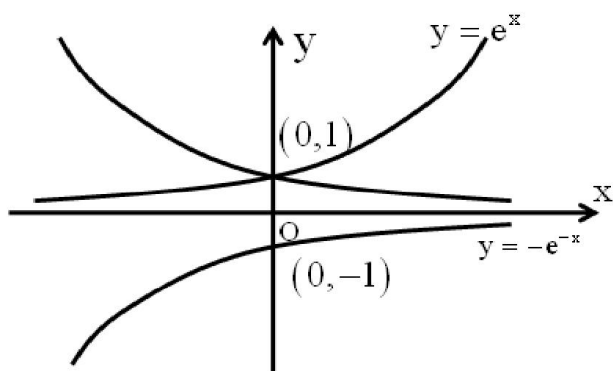
known graph of $y = f(x)$

Since $f(-x)$ is symmetrical to that of $f(x)$ about the y-axis and $-f(-x)$ is symmetrical to that of $f(-x)$ about the x-axis.

i.e., first image of $f(x)$ about y-axis later its image about the x-axis

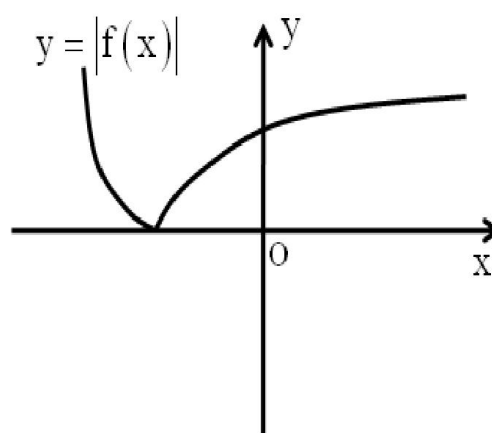
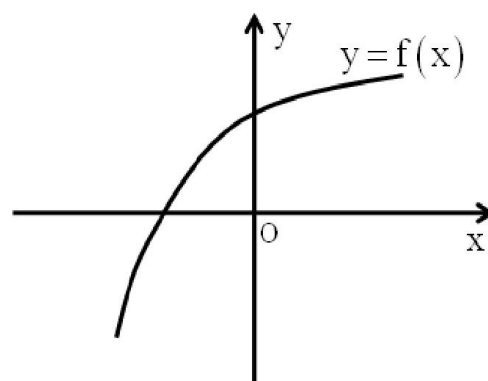


W.E-29: Drawn the graph of $y = -e^{-x}$ with the help of the known graph $y = e^x$

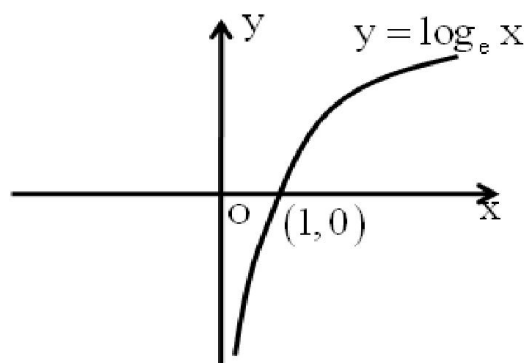


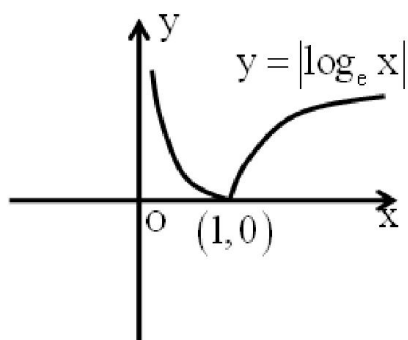
8. Drawing the graph of $y = |f(x)|$ from the known graph of $y = f(x)$

The graph of $y = |f(x)|$ is obtained from the graph of $y = f(x)$, the portion lying above the x-axis remains unchanged. Now take the image of negative part of $f(x)$ (i.e., the portion below the x-axis) in the x-axis as plane mirror.

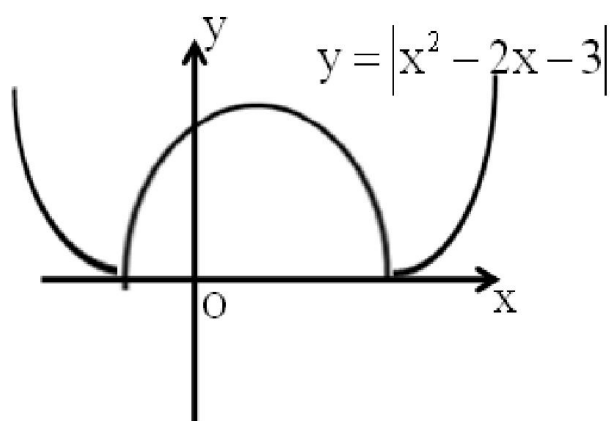
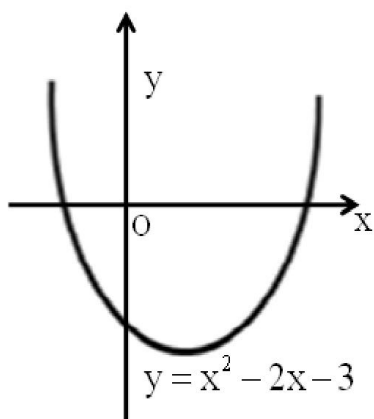


W.E-30: Draw the graph of $y = |\log_e x|$ with the help of the known function $y = \log_e x$.





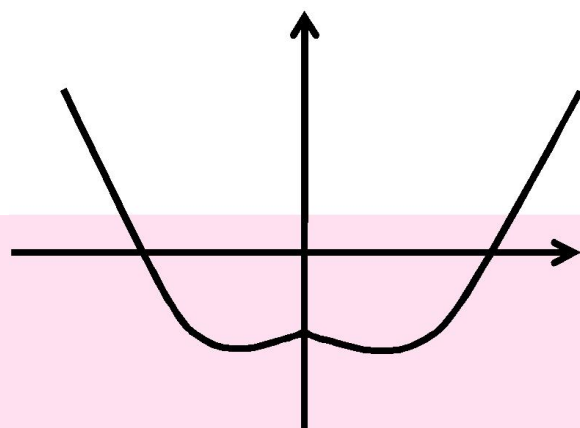
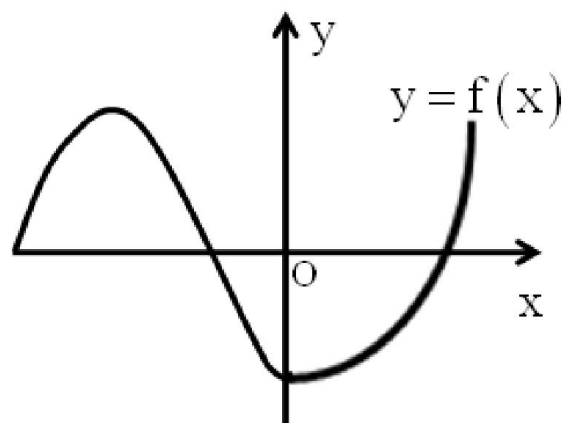
W.E-31: Draw the graph of $y = |x^2 - 2x - 3|$ with the help of $y = x^2 - 2x - 3$



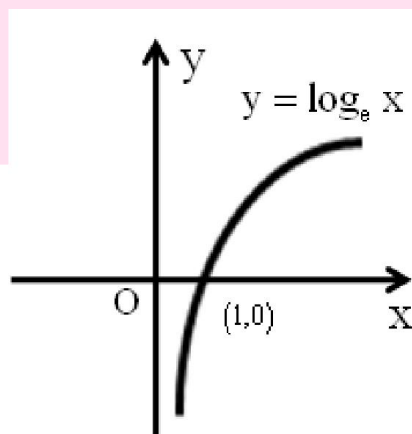
9. Drawing the graph of $y = f(|x|)$ from the known graph of $y = f(x)$

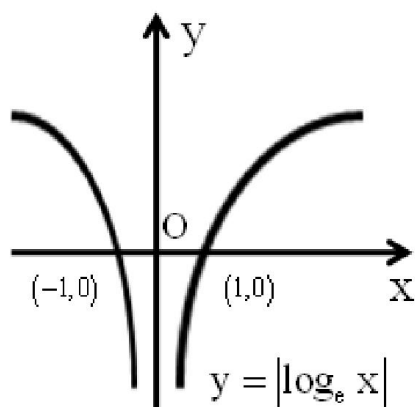
First draw the graph of $y = f(x)$ leave the graph lying right side of the y-axis as it is and also take its image w.r.t y-axis the portion of $y = f(x)$ in

the left of y-axis is ignored.



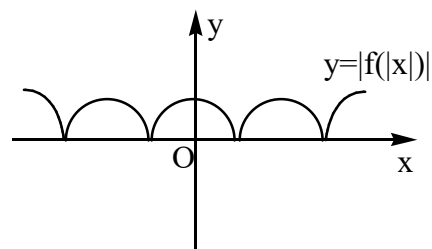
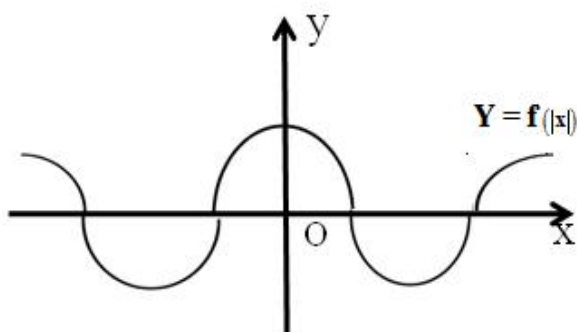
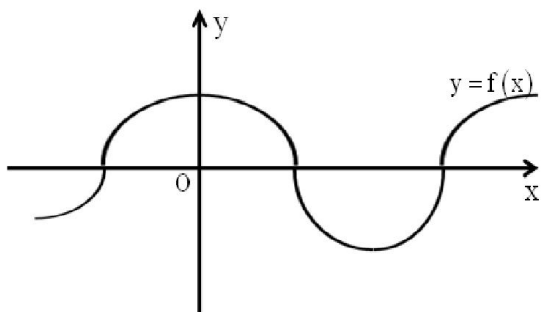
W.E-32: Sketch the graph of $y = \log_e |x|$ with the help of $y = \log_e x$.



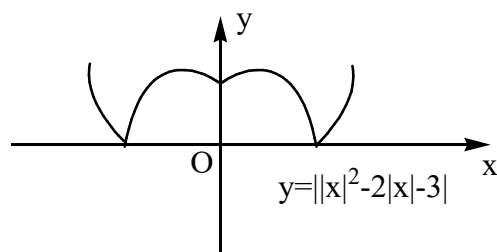
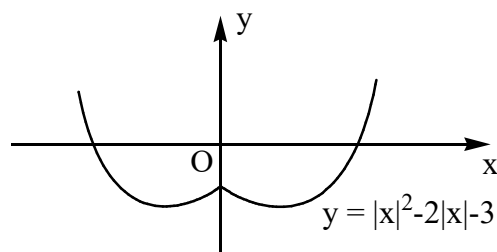
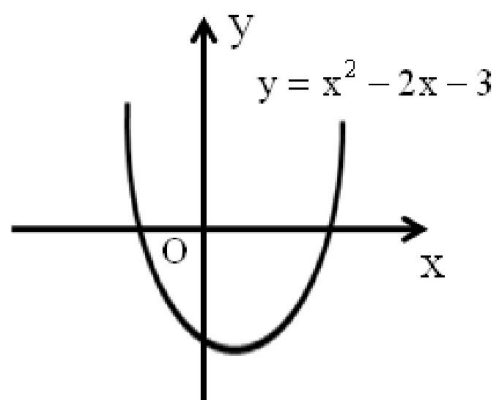


10. Drawing the graph of $y = |f(|x|)|$ from the known graph of $y = f(x)$

Here first we draw the graph of $y = f(|x|)$ from the previous method, later the part above the x-axis remains unchanged and the part below the x-axis will be shifted to the above the x-axis (i.e., take its image w.r.t x-axis)



W.E-33: Sketch the graph of $y = ||x|^2 - 2|x| - 3|$ with the help of $y = x^2 - 2x - 3$

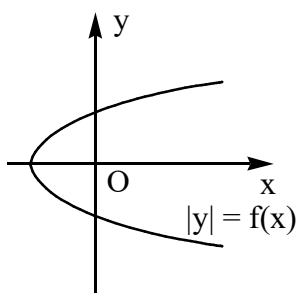
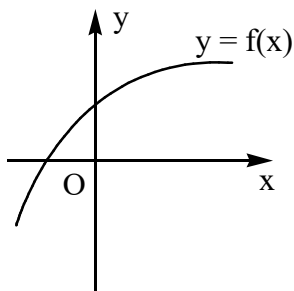


11. Drawing the graph of $|y| = f(x)$ from the known graph of $y = f(x)$

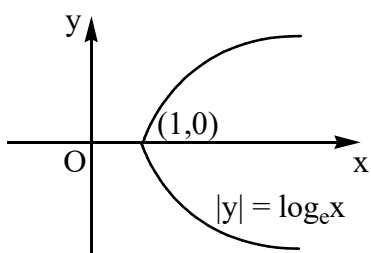
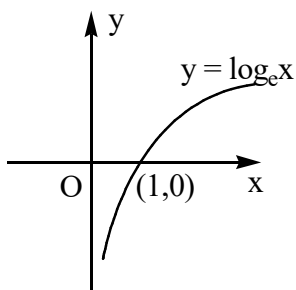
The graph of $|y| = f(x)$ is obtained from

$y = f(x)$, the portion below the x-axis is ignored, the portion above the x-axis will be same and also have to take its image w.r.t x-axis

(Here $f(x) < 0$ is not possible)

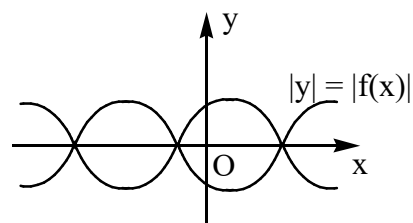
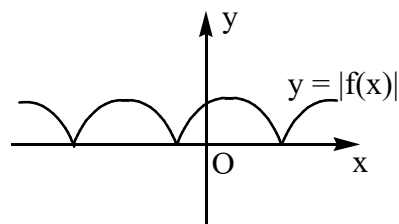
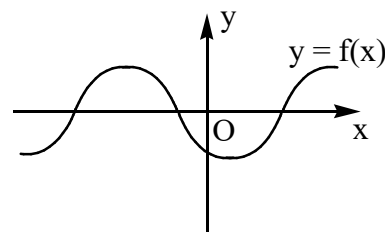


W.E-34: Draw the graph of $|y| = \log_e x$ with the help of $y = \log_e x$

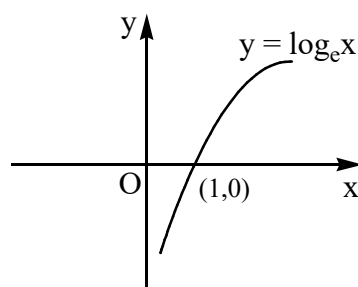


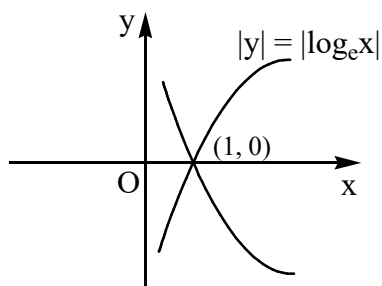
12. Drawing the graph of $|y| = |f(x)|$ from the known graph of $y = f(x)$

First we draw $|y| = f(x)$ from the previous method, then the graph of $|y| = |f(x)|$ is the portion of graph $|y| = f(x)$ lying above the x-axis remains unchanged. Its other portion located below the x-axis then take its image w.r.t x-axis



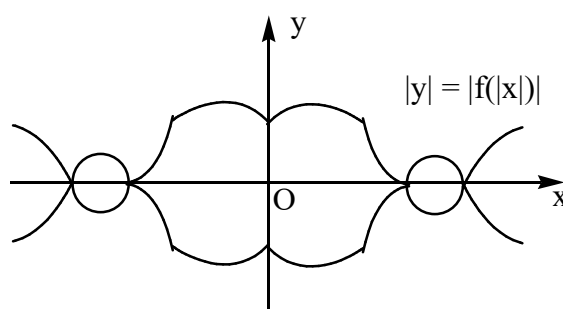
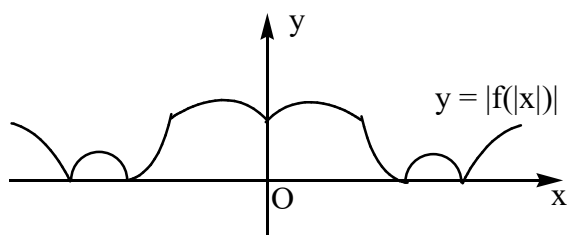
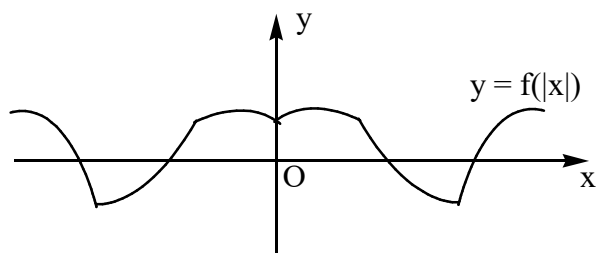
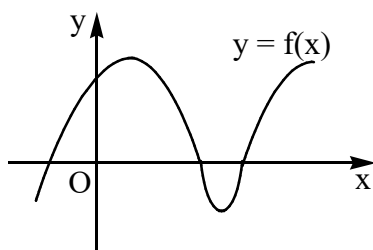
W.E-35: Draw the graph of $|y| = |\log_e x|$ from the graph of $y = \log_e x$



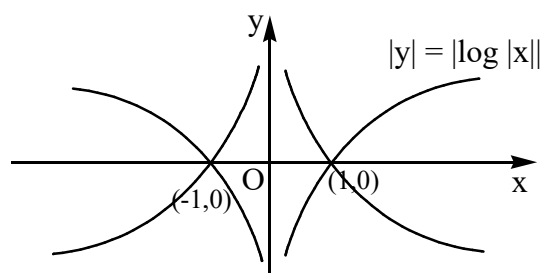
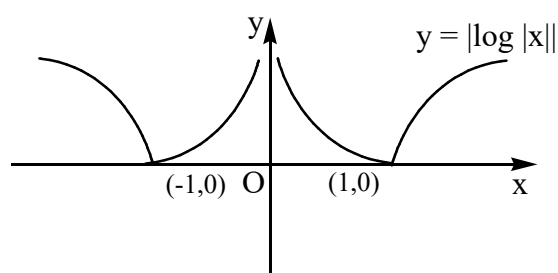
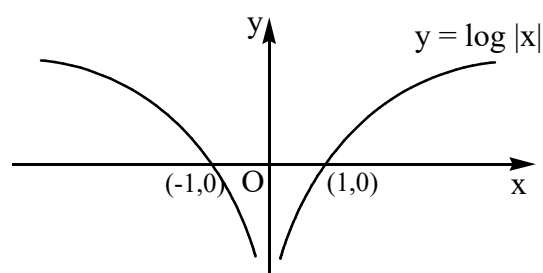
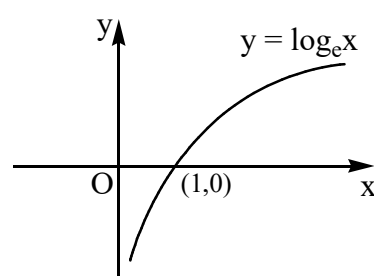


- 13. Drawing the graph of $|y| = |f(|x|)|$ from the known graph of $y = f(x)$**

The graph of $|y| = |f(|x|)|$ is obtained from the graph of $y = f(x)$. First draw the graph of $y = |f(|x|)|$ from the previous technique take its image w.r.t x-axis to get the graph of $|y| = |f(|x|)|$

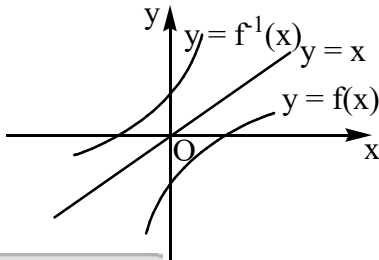


- W.E-36: Draw the graph of $|y| = |\log |x||$ with the help of $y = \log x$**



14. Drawing the graph of $y = f^{-1}(x)$ from the known graph of $y = f(x)$

The graph of $y = f^{-1}(x)$ is obtained by taking the reflection of $y = f(x)$ w.r.t the mirror line $y = x$



C.U.Q

1. $f : A \rightarrow B$ is a function then **A, B** are respectively.
 - 1) domain, range
 - 2) domain, co domain
 - 3) co domain, range
 - 4) range, domain
2. $f : A \rightarrow B$ then **$f(A)$** is called
 - 1) domain
 - 2) co domain
 - 3) range
 - 4) function
3. If $f : A \rightarrow B$ is a function then
 - 1) $f(A) = B$
 - 2) $f(A) \subset B$
 - 3) $f(A) \subseteq B$
 - 4) $B \subseteq f(A)$
4. If $f : A \rightarrow B$ is surjective then
 - 1) No two elements of A have the same image in B
 - 2) Every element in A has an image in B
 - 3) Every element of B has at least one pre-image in A
 - 4) A and B are finite non empty sets
5. A constant function $f : A \rightarrow B$ will be one-one if
 - 1) $n(A) = n(B)$
 - 2) $n(A) = 1$
 - 3) $n(B) = 1$
 - 4) $n(A) < n(B)$
6. If $f : A \rightarrow B$ is a constant function which is onto then B is
 - 1) a singleton set
 - 2) a null set
 - 3) an infinite set
 - 4) a finite set
7. If $n \geq 2$ then the number of surjections that can be defined from $\{1, 2, 3, \dots, n\}$ onto $\{1, 2\}$ is
 - 1) $n^2 - n$
 - 2) n^2
 - 3) 2^n
 - 4) $2^n - 2$
8. If f and g are functions such that $f \circ g$ is onto then
 - 1) f is onto
 - 2) g is onto
 - 3) $g \circ f$ is onto
 - 4) Neither f nor g is onto
9. If f and g are functions such that $f \circ g$ is one - one then
 - 1) g must be onto
 - 2) g must be one - one
 - 3) f must be one-one
 - 4) f, g need not one-one
10. To have inverse for the function f, f should be
 - 1) one-one
 - 2) onto
 - 3) one-one and onto
 - 4) Identity function
11. If $f : A \rightarrow B$ is a bijection then $f^{-1} \circ f =$
 - 1) $f \circ f^{-1}$
 - 2) f
 - 3) f^{-1}
 - 4) I_A
12. $f : A \rightarrow B$ is a bijection then $f \circ f^{-1} =$
 - 1) I_A
 - 2) I_B
 - 3) f
 - 4) f^{-1}
13. Let $f(x) = ax^2 + bx + c$, where a, b, c are rational and $f : Z \rightarrow Z$ where Z is the set of integers. Then $a + b$ is
 - 1) a negative integer
 - 2) an integer
 - 3) non integral rational number
 - 4) Real number
14. The graph of the function $y = f(x)$ is symmetrical about the line $x = 2$. Then
 - 1) $f(x+2) = f(x-2)$
 - 2) $f(2+x) = f(2-x)$
 - 3) $f(x) = f(-x)$
 - 4) $f(x) = -f(-x)$
15. The graph of odd function is
 - 1) symmetric about origin
 - 2) symmetric about y-axis
 - 3) parallel to x-axis
 - 4) parallel to y-axis
16. The graph of even function is
 - 1) symmetric in opp quadrants
 - 2) symmetric about y-axis
 - 3) parallel to x-axis
 - 4) parallel to y-axis
17. The graph of Identify function is
 - 1) Straight line passing through origin
 - 2) symmetric about y-axis
 - 3) parallel to x-axis
 - 4) parallel to y-axis
18. The graph of $|x|$ is
 - 1) Straight line passing through origin
 - 2) symmetric about y-axis

3) parallel to x-axis 4) parallel to y-axis

19. The function $f(x) = \{x\}$ is

- 1) even function 2) odd function
3) periodic function with period 1
4) periodic function but no fundamental period

20. Let f be a strictly decreasing function with range $[a, b]$ then domain of the function f^{-1} is

- 1) $[f^{-1}(b), f^{-1}(a)]$ 2) $[b, a]$
3) $[f^{-1}(a), f^{-1}(b)]$ 4) (b, a)

21. If $f(x) = x^2 + \lambda x + \mu$ be an integral function of the integral variable x then

- 1) λ is an integer and μ is a rational fraction
2) λ and μ are integers
3) μ is an integer and λ is a rational fraction
4) λ and μ are rational fractions

C.U.Q-KEY

- 01) 2 02) 3 03) 3 04) 3 05) 2
06) 1 07) 4 08) 1 09) 2 10) 3
11) 4 12) 2 13) 2 14) 2 15) 1
16) 2 17) 1 18) 2 19) 3 20) 1
21) 2



LEVEL - I (C.W)

Real valued functions :

1. Let $f: \{(1,1), (2,3), (0,-1), (-1,-3)\}$ be a function from z to z defined by $f(x) = ax+b$, for some integers a, b then $(a, b) =$

- 1) $(-1, 2)$ 2) $(2, -1)$ 3) $(3, -2)$ 4) $(0, 3)$

2. If $e^{f(x)} = \frac{10+x}{10-x}, x \in (-10, 10)$

and $f(x) = k \cdot f\left(\frac{200x}{100+x^2}\right)$ then $k =$

- 1) 0.5 2) 0.6 3) 0.7 4) 0.8

3. $f: R \rightarrow R$ is defined as $f(x) = 2x + |x|$ then $f(3x) - f(-x) - 4x =$ (EAM-97)

- 1) $f(x)$ 2) $-f(x)$ 3) $f(-x)$ 4) $2f(x)$

4. $f(1) = 1, n \geq 1 \Rightarrow f(n+1) = 2f(n) + 1$ then $f(n) =$ (EAM-92)

- 1) 2^{n+1} 2) 2^n 3) $2^n - 1$ 4) $2^{n-1} - 1$

5. If $f(x) = \frac{7^{1+\ln x}}{x^{\ln 7}}$ then $f(2015) =$

- 1) 20 2) 7 3) 2015 4) 100

6. If $f(x) = \frac{\cos^2 x + \sin^4 x}{\sin^2 x + \cos^4 x}$ for $x \in R$ then

$f(2016) =$

- 1) 1 2) 2 3) 3 4) 4

7. If $f = \{(-2, 4), (0, 6), (2, 8)\}$ and

$g = \{(-2, -1), (0, 3), (2, 5)\}$, then

$$\left(\frac{2f}{3g} + \frac{3g}{2f}\right)(0) =$$

- (1) $1/12$ (2) $25/12$ (3) $5/12$ (4) $13/12$

8. If $f(x) = \sin(\log x)$ then

$$f(xy) + f\left(\frac{x}{y}\right) - 2f(x) \cos(\log y) =$$

- 1) $\cos(\log x)$ 2) $\sin(\log y)$
3) $\cos(\log(xy))$ 4) 0

9. If $f(x+y, x-y) = xy$ then the arithmetic mean of $f(x, y)$ and $f(y, x)$ is

- 1) x 2) y 3) 0 4) xy

Even and odd functions :

10. Let $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$, then f is

- 1) an odd function 2) an even function
3) both odd and even 4) neither odd nor even

11. Which of the following is an even function

1) $f(x) = \frac{a^x + a^{-x}}{a^x - a^{-x}}$ 2) $f(x) = \frac{a^x + 1}{a^x - 1}$

3) $f(x) = x \frac{a^x - 1}{a^x + 1}$

4) $f(x) = \log_2(x + \sqrt{x^2 + 1})$

Periodic functions:

12. The period of $\cos x^2$ is
1) 2π 2) $\sqrt{2}\pi$ 3) $4\pi^2$ 4) does not exist
13. Period of $f(x) = e^{\cos\{x\}} + \sin \pi[x]$ is $([\cdot])$ and $\{\cdot\}$ denote the greatest integer function and fractional part function respectively)
1) 1 2) 2 3) π 4) 2π
14. Let $f(x)$ be periodic and k be a positive real number such that $f(x+k) + f(x) = 0$ for all $x \in \mathbb{R}$. Then the period of $f(x)$ is
1) k 2) $2k$ 3) $4k$ 4) $8k$
15. The period of $f(x) = \sqrt{x - [x]}$ is
1) no fundamental period 2) $\frac{1}{2}$
3) 1 4) 2

Domain of the function:

16. The domain of $f(x) = \frac{x^2 + 2x + 1}{x^2 - x - 6}$
1) $\mathbb{R} - \{3, -2\}$ 2) $\mathbb{R} - \{-3, 2\}$
3) $\mathbb{R} - \{3, -2\}$ 4) $\mathbb{R} - (3, -2)$
17. The domain of $f(x) = \frac{1}{\log|x|}$ is
1) $\mathbb{R} - \{0\}$ 2) $\mathbb{R} - \{0, 1\}$
3) $\mathbb{R} - \{-1, 0, 1\}$ 4) $(-\infty, \infty)$
18. The domain of $f(x) = \frac{3^x + 3^{-x}}{3^x - 3^{-x}}$ is
1) $(-\infty, \infty)$ 2) $(-\infty, 0) \cup (0, \infty)$
3) $(0, \infty)$ 4) $(0, 1)$
19. The domain of the function $f(x) = \sqrt{\log_{16} x^2}$ is
1) $x = 0$ 2) $|x| \geq 4$ 3) $|x| \geq 1$ 4) $|x| \geq 2$

20. The domain of $f(x) = \frac{1}{\sqrt{|x| - x}}$ is [AIE-2011]
1) $(-\infty, 0)$ 2) $(0, \infty)$ 3) $(1, \infty)$ 4) $(-\infty, \infty)$
21. The domain of $f(x) = \frac{1}{[x] - x}$ is
1) $\mathbb{R}$ 2) $\mathbb{Z}$ 3) $\mathbb{R} - \mathbb{Z}$ 4) $\mathbb{Q} - \{0\}$
22. The domain of $f(x) = \sqrt{x-2} + \frac{1}{\log(4-x)}$ is
1) $[2, \infty)$ 2) $(-\infty, 4)$
3) $[2, 3) \cup (3, 4)$ 4) $[3, \infty)$
23. The domain of $f(x) = e^{\sqrt{x}} + \cos x$ is
1) $(-\infty, \infty)$ 2) $[0, \infty)$ 3) $(0, 1)$ 4) $(1, \infty)$
24. The domain of $\log_a \sin^{-1} x$ is ($a > 0, a \neq 1$)
1) $0 < x \leq 1$ 2) $0 \leq x \leq 1$
3) $0 \leq x < 1$ 4) $0 < x < 1$
25. The domain of $\cosh^{-1} 5x$ is
1) $\mathbb{R}$ 2) $[0, \infty)$ 3) $\left(\frac{1}{5}, \infty\right)$ 4) $\left[\frac{1}{5}, \infty\right)$
26. For which Domain, the functions $f(x) = 2x^2 - 1$ and $g(x) = 1 - 3x$ are equal to
1) $\mathbb{R}$ 2) $\left\{\frac{1}{2}, -2\right\}$ 3) $\left(\frac{1}{2}, 2\right)$ 4) $\left[\frac{1}{2}, 2\right]$

Range of the function :

27. The domain and range of the real function f defined by $f(x) = \frac{4-x}{x-4}$ is given by
1) Domain = $\mathbb{R}$, Range = $\{-1, 1\}$
2) Domain = $\mathbb{R} - \{1\}$, Range = $\mathbb{R}$
3) Domain = $\mathbb{R} - \{4\}$, Range = $\{-1\}$
4) Domain = $\mathbb{R} - \{-4\}$, Range = $\{-1, 1\}$
28. Range of $f(x) = \frac{1}{1 - 2 \cos x}$ is
1) $\left[\frac{1}{3}, 1\right]$ 2) $\left[-1, \frac{1}{3}\right]$
3) $(-\infty, -1] \cup \left[\frac{1}{3}, \infty\right)$ 4) $\left[-\frac{1}{3}, 1\right]$

29. The range of $f(x) = x^2 + x + 1$ is

- 1) $\left[\frac{3}{4}, \infty\right)$ 2) $[0, \infty)$ 3) $[1, \infty)$ 4) $\left[\frac{1}{4}, \infty\right)$

30. The domain and range of the function f given by $f(x) = 2 - |x - 5|$ is

- 1) Domain = $\mathbb{R}^+$, Range = $(-\infty, 1]$
 2) Domain = $\mathbb{R}$, Range = $(-\infty, 2]$
 3) Domain = $\mathbb{R}$, Range = $(-\infty, 2)$
 4) Domain = $\mathbb{R}^+$, Range = $(-\infty, 2]$

31. $f = \left\{ \left(x, \frac{x^2}{x^2 + 1} \right) : x \in \mathbb{R} \right\}$, be a function $\mathbb{R}$ into $\mathbb{R}$, range of 'f'

- 1) $[0, 1)$ 2) $(-\infty, \infty)$ 3) $(0, \infty)$ 4) $\mathbb{R}^+$

32. Range of the function $f(x) = \sqrt{[x]} - x$ is

- 1) $\mathbb{R}$ 2) $\{1\}$ 3) $\{0\}$ 4) $(0, \infty)$

33. Let $A = \{9, 10, 11, 12, 13\}$ and $f: A \rightarrow \mathbb{N}$ be defined by $f(n) =$ highest prime factor of n , then its range is

- 1) $\{13\}$ 2) $\{3, 5, 11, 13\}$
 3) $\{11, 13\}$ 4) $\{2, 3, 5, 11\}$

34. The range of $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$ is

- 1) $\left[\frac{1}{3}, 3\right]$ 2) $\left[\frac{1}{2}, 2\right]$ 3) $[0, 1]$ 4) $[-1, 1]$

Types of functions :

35. The equivalent function of $\log x^2$ is

- 1) $2 \log x$ 2) $2 \log |x|$
 3) $|\log x^2|$ 4) $(\log x)^2$

36. The number of linear functions which map $[-1, 1]$ to $[0, 2]$ are

- 1) One 2) Two 3) Four 4) Three

37. If $A = (3, 81)$ and $f: A \rightarrow B$ is a surjection defined by $f(x) = \log_3 x$ then $B =$

- 1) $[1, 4]$ 2) $(1, 4]$ 3) $(1, 4)$ 4) $[1, \infty)$

38. Let $f(x) = \sin^2 \frac{x}{2} + \cos^2 \frac{x}{2}$ and

$g(x) = \sec^2 x - \tan^2 x$. The two functions are equal over the set

- 1) ϕ 2) $\mathbb{R}$
 3) $\mathbb{R} - \left\{ x \mid x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z} \right\}$ 4) $\mathbb{R} - \{0\}$

39. $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{x^2 + 1}$, $\forall x \in \mathbb{R}$ is

- 1) one - one 2) onto
 3) bijective 4) neither one one nor onto

40. If $f: \mathbb{Z} \rightarrow \mathbb{Z}$ is such that $f(x) = 6x - 11$ then f is

- 1) injective but not surjective
 2) surjective but not injective 3) bijective
 4) neither injective nor surjective

41. $f: \mathbb{N} \rightarrow A$ Where $A = \{0, 1\}$ defined by

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is odd} \\ 1 & \text{if } x \text{ is even} \end{cases} \text{ Then } f \text{ is}$$

- (1) one - one, onto (2) one-one, into
 (3) many-one, onto (4) many-one, into

42. $f: (-\infty, \infty) \rightarrow (0, 1]$ defined by

$$f(x) = \frac{1}{x^2 + 1} \text{ is}$$

- 1) one-one but not onto
 2) onto but not one-one
 3) bijective 4) neither one-one nor onto

43. The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = 4^x + 4^{|x|} \text{ is}$$

- 1) One - one and into 2) Many - one and into
 3) One - one and onto 4) Many-one and onto

Number of functions :

44. The number of one-one functions that can be defined from $A = \{4, 8, 12, 16\}$ to B is 5040, then $n(B) =$

- 1) 7 2) 8 3) 9 4) 10

45. If $A = \{1, 8, 11, 14, 25\}$ then the condition to define a surjection from A to B is

- 1) $n(A) + n(B) = 20$ 2) $n(A) < n(B)$
 3) $n(B) \leq 5$ 4) $n(B) = 10$

46. If $A = \{1, 2, 3\}$, $B = \{1, 2\}$ then the number of functions from A to B are
1) 6 2) 8 3) 9 4) 32
47. The number of non-bijective mappings that can be defined from $A = \{1, 2, 7\}$ to itself is
1) 21 2) 27 3) 6 4) 9
48. Let $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$. If l is number of functions from A to B and m is number of one-one functions from A to B, then
1) l is 9 2) m is 9 3) l is 27 4) m is 16
49. The number of constant functions possible from R to B where $B = \{2, 4, 6, 8, \dots, 24\}$ are
1) 24 2) 12 3) 8 4) 6

Composite functions :

50. The functions $f: R \rightarrow R$, $g: R \rightarrow R$ are defined as

$$f(x) = \begin{cases} 0 & \text{when } x \text{ is rational} \\ 1 & \text{when } x \text{ is irrational} \end{cases}$$

$$g(x) = \begin{cases} -1 & \text{when } x \text{ is rational} \\ 0 & \text{when } x \text{ is irrational} \end{cases}$$

then $(f \circ g)(\pi) + (g \circ f)(e) =$ (EAM-01)

- 1) -1 2) 0 3) 1 4) 2

51. If $f(x) = (a - x^n)^{\frac{1}{n}}$ then $f \circ f(x)$ is

- 1) x 2) $a - x$ 3) x^n 4) $\frac{-1}{x^n}$

52. If $f(x) = \frac{x}{\sqrt{1+x^2}}$ then $f \circ f \circ f(x) =$

- 1) $\frac{x}{\sqrt{1+3x^2}}$ 2) $\frac{x}{\sqrt{1-x^2}}$
3) $\frac{2x}{\sqrt{1+2x^2}}$ 4) $\frac{x}{\sqrt{1+x^2}}$

Inverse of a function:

53. If $f(x) = \frac{e^x + e^{-x}}{2}$ then the inverse of $f(x)$ is

- 1) $\log_e(x + \sqrt{x^2 + 1})$ 2) $\log_e \sqrt{x^2 + 1}$
3) $\log_e(x + \sqrt{x^2 - 1})$ 4) $\log_e(x - \sqrt{x^2 - 1})$

54. If $f: \{1, 2, 3, \dots\} \rightarrow \{0, \pm 1, \pm 2, \dots\}$ is defined

$$\text{by } f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ -\left(\frac{n-1}{2}\right) & \text{if } n \text{ is odd} \end{cases}$$

then $f^{-1}(-100)$ is

- 1) 100 2) 199 3) 201 4) 200

55. If $f(x) = \sin^{-1}\{3 - (x-6)^4\}^{1/3}$ then $f^{-1}(x) =$

- 1) $6 + \sqrt[4]{3 + \sin^3 x}$ 2) $6 + \sqrt[4]{3 - \sin^3 x}$
3) $6 + \sqrt[4]{3 + \sin x}$ 4) $6 + \sqrt[4]{3 - \sin x}$

LEVEL-I(C.W.)-KEY

- 01) 2 02) 1 03) 4 04) 3 05) 2
06) 1 07) 2 08) 4 09) 3 10) 2
11) 3 12) 4 13) 1 14) 2 15) 3
16) 1 17) 3 18) 2 19) 3 20) 1
21) 3 22) 3 23) 2 24) 1 25) 4
26) 2 27) 3 28) 3 29) 1 30) 2
31) 1 32) 3 33) 2 34) 1 35) 2
36) 2 37) 3 38) 3 39) 4 40) 1
41) 3 42) 2 43) 1 44) 4 45) 3
46) 2 47) 1 48) 3 49) 2 50) 1
51) 1 52) 1 53) 3 54) 3 55) 2

LEVEL-I(C.W.)-HINTS

- solve the equations
- $f(x) = 2k f(x)$
- expand and it is $2 f(x)$
- Put $n = 1$
- $7^{\ell n x} = x^{\ell n 7}$
- $f(x) = \frac{\cos^2 x + \sin^2 x (1 - \cos^2 x)}{\sin^2 x + \cos^2 x (1 - \sin^2 x)} = 1$
- $\left(\frac{2f}{3g} + \frac{3g}{2f}\right)(0) = \frac{2f(0)}{3g(0)} + \frac{3g(0)}{2f(0)} = \frac{25}{12}$
- $\sin(\log x + \log y) + \sin(\log x - \log y) - 2\sin(\log x)\cos(\log y) = 0$
- Let $x + y = p$, $x - y = q$. Then
$$f(p, q) = \frac{p+q}{2} \cdot \frac{p-q}{2} = \frac{p^2 - q^2}{4}$$

$$\therefore f(x, y) = \frac{x^2 - y^2}{4} \text{ and } f(y, x) = \frac{y^2 - x^2}{4}$$

10. $f(-x) = f(x)$
11. verify that the options are even and odd
12. no value of T exist such that $f(x+T) = f(x)$
13. $f(x)e^{\cos\{x\}}$ period of $\{x\}$.i.e. of $x - [x]$ is 1
 $\therefore$ period of $f(x)$ is 1.
14. We have, $f(x+k) + f(x) = 0 \dots (1)$ for all $x \in R$.
 Putting $x+k$ for x , we get
 $f(x+2k) + f(x+k) = 0 \dots (2)$
 $[(2) - (1)] \Rightarrow f(x+2k) - f(x) = 0 \Rightarrow$
 $f(x+2k) = f(x)$
 This shows that $f(x)$ is periodic with period $2k$.
15. $f(x) = \sqrt{\{x\}}$ period is one
16. denominator not equal to zero
17. $\log|x| \neq 0$ and $x \neq 0 \Rightarrow x \neq \{0, 1, -1\}$
18. $3^x \neq 3^{-x}$
19. $x \neq 0$ and $x^2 \geq 1 \Rightarrow |x| \geq 1$
20. $|x| - x > 0$
21. $[x] - x \neq 0$
22. $x - 2 \geq 0, 4 - x > 0$ and $x \neq 3$
23. $x \geq 0$
24. $\sin^{-1}x > 0 \Rightarrow 0 < x \leq 1$
25. Domain of $\cosh^{-1}x = [1, \infty)$
26. $2x^2 - 1 = 1 - 3x \Rightarrow (2x-1)(x+2) = 0$
27. denominator not equal to zero.
28. $-2 \leq -2\cos x \leq 2, -1 \leq 1 - 2\cos x \leq 3$
29. $\left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4}$
30. $|x|$ defines for any value
31. $x^2 < x^2 + 1 \therefore \frac{x^2}{x^2 + 1} \in [0, 1)$
32. Domain of $x = z, \therefore [x] - x = 0 \forall x \in Z$
33. $f(9) = 3, f(10) = 5, f(11) = 11, f(12) = 3,$
- $f(13) = 13$
34. $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1} = y$
 $\Rightarrow (y-1)x^2 + (y+1)x + (y-1) = 0$ and x is real
35. As $\log x$ is defined for only positive values of x .
 But $\log x^2$ defined for all non zero real values of x , also $\log|x|$ is defined $\forall$ non zero real x . Hence
 $\log x^2$ and $2\log|x|$ are identical functions.
36. Width of both interval is same, which can be mapped by these functions $y = 1 - x$ and $y = 1 + x$.
37. $3 < x < 81 \Rightarrow \log_3 3 < \log_3 x < \log_3 81$
38. Common domain
39. $f(2) = f\left(\frac{1}{2}\right), f(1)$ has no pre image.
40. linear function is one one
 no pre image for $y = 0$
41. $f(1) = f(3) = 0 \Rightarrow f$ is not one-one
 Range of $f = \{0, 1\} \Rightarrow f$ is onto
42. $f(-1) = f(1) \Rightarrow f$ is not one one
 range = codomain $\Rightarrow f$ is onto
43. Since for different $x, 4^x$ and $4^{|x|}$ are different positive numbers
 $\therefore f$ is one one also, f is not onto as its range $(0, \infty)$ is a proper subset of its co domain.
44. No. of one-one functions from A to $B = {}_{n(B)}P_4$
45. $n(A) \geq n(B) \Rightarrow n(B) \leq 5$
46. $n(B)^{n(A)}$
47. $3^3 - 3!$
48. The no. of functions from A to $B = 3^3 = 27$
49. The no. of constant functions from A to B is $n(B)$.
50. $(f \circ g)(\pi) + g \circ f(e) = f[g(\pi)] + g[f(e)]$
 $= f(0) + g(1) = 0 - 1 = -1$
51. $f \circ f(x) = f[f(x)] = f\left[\left(a - x^n\right)^{\frac{1}{n}}\right] = x$

52. $f(f(\dots\dots n \text{ times})) = \frac{x}{\sqrt{1+nx^2}}$

53. let $f(x) = y$

54. Take $f^{-1}(x) = y$

$$\Rightarrow x = f(y) = 5y - 6 \Rightarrow y = f^{-1}(x) = \frac{x+6}{5}$$

$$\text{Let } f^{-1}(-100) = x \Rightarrow f(x) = -100$$

$$\Rightarrow \frac{-(x-1)}{2} = -100 \Rightarrow x = 201$$

55. $f^{-1}(x) = y \Rightarrow x = f(y) = \sin^{-1}(3 - (y-6)^4)^{1/3}$

$$\Rightarrow f^{-1}(x) = 6 + \sqrt[4]{3 - \sin^3 x}.$$



LEVEL - I (H.W)

Real valued functions:

1. **f** is defined by $f(x) = \begin{cases} x^2, 0 \leq x \leq 2 \\ 3x, 2 \leq x \leq 10 \end{cases}$ then $f(2) =$

1) 9 2) 6 3) 5 4) not defined

2. If $f(x) = ax + b$, where a and b are integers, $f(-1) = -5$ and $f(3) = 3$, then a and b are equal to

1) $a = -3, b = -1$ 2) $a = 2, b = -3$
3) $a = 0, b = 2$ 4) $a = 2, b = 3$

3. If $f(x) = \begin{cases} x^2 + 1, x \leq 0 \\ 2x - 1, 0 < x < 5 \\ 4x + 3, x \geq 5 \end{cases}$

then $\frac{f(-3) + f(2) + f(5)}{f(1)} =$

1) 28 2) 36 3) 26 4) 34

4. If $f = \{(-1, 3), (0, 2), (1, 1)\}$ then the range of $f^2 - 1$ is

1) $\{0, 8\}$ 2) $\{0, 3, 8\}$ 3) $\{0, 1, 3\}$ 4) $\{0, 2, 8\}$

5. $f: R \rightarrow R$ is a function defined by $f(x) = \frac{1}{2}$

then $f\left(-\frac{2}{5}\right) =$

1) 2 2) $-\frac{1}{2}$ 3) -2 4) $\frac{1}{2}$

6. If $f(x) = \sin\left(\frac{\pi}{3}[x] - x^2\right)$ then the value of

$f\left(\sqrt{\frac{\pi}{3}}\right)$ is

1) 1 2) -1 3) 0 4) $-\frac{3}{4}$

7. The function $y = f(x)$ such that $f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}$

1) $2 - x^2$ 2) $x^2 - 2$ 3) $x^2 + 4$ 4) $4x^2 - 2$

8. If $f = \{(a, 1), (b, -2), (c, 3)\}$,

$g = \{(a, -2), (b, 0), (c, 1)\}$

then $f^2 + g^2 =$

1) $\{(a, -1), (b, -2), (c, 4)\}$

2) $\{(a, 3), (b, -2), (c, 2)\}$

3) $\{(a, -4), (b, -4), (c, 9)\}$

4) $\{(a, 5), (b, 4), (c, 10)\}$

9. Which of the following relations are functions

$f: \{(2, 1), (3, 1), (4, 2)\}$

$g: \{(2, 2), (2, 4), (3, 3), (4, 4)\}$

$h: \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$

1) f, g 2) g, h 3) h, f 4) f, g, h

Even and odd functions

10. If $f(x) = ax^5 + bx^3 + cx + d$ is odd then

1) $a = 0$ 2) $b = 0$ 3) $c = 0$ 4) $d = 0$

11. A function whose graph is symmetrical about the y axis is given by

1) $f(x) = \sin\left[\log\left(x + \sqrt{x^2 + 1}\right)\right]$

2) $f(x) = \frac{\sec^4 x + \cos^4 x}{x^3 + x^4 \cot x}$

3) $f(x + y) = f(x) + f(y) \forall x, y \in R$

4) $f(x) = x^2$

Periodic functions :12. The period of $\operatorname{sgn}(x - [x])$ is

- 1) any real number 2) 0 3) 1
4) non periodic function

13. The period of $x \cos x$ is

- 1) 2π 2) π 3) $\frac{\pi}{2}$ 4) non periodic

Domain of the function:14. Domain of $f(x) = \frac{x^2 + 3x + 5}{x^2 - 5x + 4}$ is

- 1) $\mathbb{R} - \{1, 4\}$ 2) $\{1, 4\}$ 3) $(1, 4)$ 4) $(1, 4]$

15. Domain of $[x] + x$ is

- 1) $\mathbb{R}$ 2) $\mathbb{Z}$ 3) $\mathbb{R} - \mathbb{Z}$ 4) $\mathbb{Q}$

16. Domain of $|x - 1|$ is

- 1) $[1, \infty)$ 2) $\mathbb{R}$ 3) $[0, \infty)$ 4) $\mathbb{Z}$

17. The domain of $f(x) = \sqrt{-x^2}$ is

- 1) $(0, \infty)$ 2) $(-\infty, 0)$ 3) $\{0\}$ 4) $(1, \infty)$

18. The domain of $f(x) = \frac{1}{|x| + x}$ is

- 1) $(-\infty, 0)$ 2) $(0, \infty)$ 3) $(-\infty, 1)$ 4) $(-2, -1)$

19. The domain of $f(x) = \sqrt{1 - |x|}$ is

- 1) $[-1, 1]$ 2) $(-1, 1)$ 3) $(0, 1)$ 4) $\mathbb{R}$

20. The domain of $f(x) = \log \{(x-3)(6-x)\}$ is

- 1) $(3, \infty)$ 2) $(3, 6)$ 3) $(0, \infty)$ 4) $(-\infty, \infty)$

21. The domain of $f(x) = \cot \frac{x}{3}$ is

- 1) $(-\infty, \infty)$ 2) $\mathbb{R} - \{n\pi : n \in \mathbb{Z}\}$
3) $\mathbb{R} - \{3n\pi : n \in \mathbb{Z}\}$ 4) $(0, \infty)$

22. The domain of $f(x) = \tan^{-1}(5x)$ is

- 1) $(-\infty, \infty)$ 2) $(0, \infty)$ 3) $(-\infty, 0)$ 4) $\left(-\frac{1}{5}, \frac{1}{5}\right)$

23. The domain of $f(x) = \sqrt[3]{x} \cot x$ is

- 1) $\mathbb{R}$ 2) $\mathbb{R} - \{n\pi : n \in \mathbb{Z}\}$

3) $\mathbb{R} - \left\{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$ 4) $(0, \infty)$

24. Domain of $f(x) = \frac{|x| - x}{2x}$ is

- 1) $\mathbb{R}$ 2) $\mathbb{R} - \{0\}$ 3) $\mathbb{Z}$ 4) $\mathbb{N}$

25. Domain of $\frac{3^x}{x+1}$ is

- 1) $\mathbb{R}$ 2) $\mathbb{R} - \{-1\}$ 3) $(1, \infty)$ 4) $(-\infty, 1)$

Range of the function:26. Range of $\frac{|x-4|}{x-4}$ is

- 1) $\mathbb{R} - \{4\}$ 2) $\mathbb{R}$ 3) $\{-1, 1\}$ 4) $\mathbb{R} - \{-1, 1\}$

27. If x is positive, the values of

- $f(x) = -3 \cos \sqrt{3+x+x^2}$ lie in the interval
1) $[-1, 3]$ 2) $[-3, 3]$ 3) $[0, 3]$ 4) $[-3, 0]$

28. The range of $f(x) = \frac{\sin \pi[x^2 - 1]}{x^4 + 1}$ is, where $[.]$ is greatest integer function

- 1) $\mathbb{R}$ 2) $[-1, 1]$ 3) $\{0, 1\}$ 4) $\{0\}$

29. The range of $f(x) = |x-2| + |x-12|$ is

- 1) $[2, \infty)$ 2) $(12, \infty)$ 3) $[10, \infty)$ 4) $[14, \infty)$

30. The range of $f(x) = 3x^2 + 7x + 10$ is

- 1) $\left[\frac{70}{3}, \infty\right)$ 2) $\left[\frac{71}{12}, \infty\right)$ 3) $[0, \infty)$ 4) $\left(-\infty, \frac{70}{3}\right)$

31. The range of the function

$$f(x) = \frac{2+x}{2-x}, x \neq 2 \text{ is } \quad \text{[AIE-2002]}$$

- 1) $\mathbb{R}$ 2) $\mathbb{R} - \{-1\}$ 3) $\mathbb{R} - \{1\}$ 4) $\mathbb{R} - \{2\}$

32. The range of $f(x) = x - [x]$ is

- 1) $x = \{1, 2, 3, \dots\}$ 2) $x \geq 0$
3) $x < 1$ 4) $0 \leq x < 1$

33. Range of $\sqrt{9-x^2}$ is

- 1) $[0, 3]$ 2) $[-3, 3]$ 3) $[-3, 0]$ 4) $\mathbb{R}$

34. Range of $f(x) = e^x$ is

- 1) $(0, \infty)$ 2) $[0, \infty)$ 3) $(-\infty, \infty)$ 4) $[e, \infty)$

Types of functions

35. If $f : R \rightarrow S$, defined by

$f(x) = \sin x - \sqrt{3} \cos x + 1$, is onto, then the interval of 'S' is [AIE-2004]

- 1) $[0, 3]$ 2) $[-1, 3]$ 3) $[0, 1]$ 4) $[-1, 1]$

36. If $f : [1, \infty) \rightarrow B$ defined by $f(x) = x^2 - 2x + 6$ is a surjection then $B =$

- 1) $[1, \infty)$ 2) $[5, \infty)$ 3) $[6, \infty)$ 4) $[2, \infty)$

37. $f : (-\infty, \infty) \rightarrow (-\infty, \infty)$ is defined by

$f(x) = ax + b, a, b \in R$ ($a \neq 0$) then f is

- 1) injective but not surjective
2) surjective but not injective 3) bijective
4) neither injective nor surjective

38. $f : Z \rightarrow Z$ and $f(x) = x^2$ then f is (EAM-95)

- 1) bijection 2) injection
3) surjection 4) not bijection

39. The function $f : R \rightarrow R$ defined by $f(x) = \sin x$ is [AIE-2002]

- 1) Neither one one nor onto 2) onto
3) one-one 4) many one

40. $f : Z \rightarrow Z$ defined as $f(x) = [x]$ then f is

- 1) not a function 2) many-to-one function
3) into function 4) identity function

41. $f : Q \rightarrow Q$ is defined by $f(x) = 15x + 7$ is

- 1) injective only 2) surjective only
3) bijective 4) neither injective nor surjective

42. $f : (0, \infty) \rightarrow [0, \infty)$ defined by $f(x) = x^2$ is

- 1) one-one but not onto 2) onto but not one-one
3) bijective 4) neither one-one nor onto

43. Let $f : [0, \infty) \rightarrow [0, 2]$ be defined by

$f(x) = \frac{2x}{1+x}$ then f is

- 1) one-one but not onto
2) onto but not one-one
3) both one-one and onto
4) neither one-one nor onto

Number of functions:

44. The total number of functions from A to itself is 256, then $n(A) =$

- 1) 2 2) 3 3) 4 4) 5

45. If $B = \{1, 2, 3\}$ and $A = \{4, 5, 6, 7, 8\}$ then the number of surjections from A to B is

- 1) 81 2) 64 3) 48 4) 150

46. The number of one-one functions that can be defined from $A = \{1, 2, 3\}$ to $B = \{a, e, i, o, u\}$ is

- 1) 3^5 2) 5^3 3) 5P_3 4) $5!$

47. The number of possible many to one functions from $A = \{6, 36\}$ to

$B = \{1, 2, 3, 4, 5\}$ is

- 1) 32 2) 25 3) 5 4) 20

48. If $n(A) = 4$ and $n(B) = 6$, then the number of surjections from A to B is

- 1) 4^6 2) 6^4 3) 0 4) 24

49. The number of bijections from the set A to itself when A contains 106 elements is

- 1) 106 2) 106^2 3) $106!$ 4) 2^{106}

50. The number of non-surjective mappings that can be defined from $A = \{1, 4, 9, 16\}$ to

$B = \{2, 8, 16, 32, 64\}$ is

- 1) 1024 2) 20 3) 505 4) 625

Composite functions:

51. If $f(x) = 2x + 1$ and $g(x) = x^2 + 1$ then

$go(fof)(2) =$

- 1) 112 2) 122 3) 12 4) 124

52. If $f(x) = [x]$, $g(x) = x - [x]$ then which of the following functions is a zero function

- 1) $(f + g)(x)$ 2) $(fg)(x)$
3) $(f - g)(x)$ 4) $(fog)(x)$

53. Let $f(x) = \frac{Kx}{x+1}$ ($x \neq -1$) then the value of K for which $(f \circ f)(x) = x$ is

- 1) 1 2) -1 3) 2 4) $\sqrt{2}$

54. If $f(x) = \frac{x}{\sqrt{1-x^2}}$, $g(x) = \frac{x}{\sqrt{1+x^2}}$ then $(fog)(x) =$

- 1) x 2) $\frac{x}{\sqrt{1+x^2}}$ 3) $\sqrt{1+x^2}$ 4) $2x$

55. If $f : [-6, 6] \rightarrow R$ is defined by

$f(x) = x^2 - 3$ for $x \in R$ then (EAM-08)

$(fofof)(-1) + (fofof)(0) + (fofof)(1) =$

- 1) $f(4\sqrt{2})$ 2) $f(3\sqrt{2})$
3) $f(2\sqrt{2})$ 4) $f(\sqrt{2})$

56. If $f(x) = (1-x)^{1/2}$ and $g(x) = \ln(x)$ then the domain of $(g \circ f)(x)$ is

1) $(-\infty, 2)$ 2) $(-1, 1)$ 3) $(-\infty, 1]$ 4) $(-\infty, 1)$

Inverse of a function:

57. If $f: R^+ \rightarrow R$ such that $f(x) = \log_5 x$ then $f^{-1}(x) =$ (EAM-97)

1) $\log_x 10$ 2) 5^x 3) 3^{-x} 4) $3^{1/x}$

58. $f: R \rightarrow R$ is a function defined by

$f(x) = 10x - 7$. If $g = f^{-1}$ then $g(x) =$

1) $\frac{1}{10x-7}$ 2) $\frac{1}{10x+7}$ 3) $\frac{x+7}{10}$ 4) $\frac{x-7}{10}$

59. $f: [0, \infty) \rightarrow [4, \infty)$ is defined by

$f(x) = x^2 + 4$ then $f^{-1}(13) =$

1) 3 2) 2 3) 1 4) 4

60. If the function $f: R \rightarrow R$ be such that

$f(x) = x - [x]$, where $[x]$ denotes the greatest integer less than or equal to x then $f^{-1}(x)$ is

1) $\frac{1}{x-[x]}$ 2) $[x] - x$

3) not defined 4) $x - [x]$

LEVEL-I(H.W.)-KEY

- | | | | | |
|-------|-------|-------|-------|-------|
| 01) 4 | 02) 2 | 03) 2 | 04) 2 | 05) 4 |
| 06) 3 | 07) 2 | 08) 4 | 09) 3 | 10) 4 |
| 11) 4 | 12) 3 | 13) 4 | 14) 1 | 15) 1 |
| 16) 2 | 17) 3 | 18) 2 | 19) 1 | 20) 2 |
| 21) 3 | 22) 1 | 23) 2 | 24) 2 | 25) 2 |
| 26) 3 | 27) 2 | 28) 4 | 29) 3 | 30) 2 |
| 31) 2 | 32) 4 | 33) 1 | 34) 1 | 35) 2 |
| 36) 2 | 37) 3 | 38) 4 | 39) 1 | 40) 4 |
| 41) 3 | 42) 1 | 43) 1 | 44) 3 | 45) 4 |
| 46) 3 | 47) 3 | 48) 3 | 49) 3 | 50) 4 |
| 51) 2 | 52) 4 | 53) 2 | 54) 1 | 55) 1 |
| 56) 4 | 57) 2 | 58) 3 | 59) 1 | 60) 3 |

LEVEL-I(H.W.)-HINTS

- $f(2)$ is not defined from the definition.
- Solve the equations
- Substitute the values
- Range of $f^2 - 1 = \{3^2 - 1, 2^2 - 1, 1^2 - 1\}$
- Constant function
- $\left[\sqrt{\pi/3}\right] = 1 \Rightarrow f(x) = \sin\left(\frac{\pi}{3} - x^2\right)$
- $f\left(x + \frac{1}{x}\right) = \left(x + \frac{1}{x}\right)^2 - 2$
- $f^2 + g^2 = \{(a, 1+4), (b, 4+0), (c, 9+1)\}$
- Concept
- $f(-x) = -f(x)$
- A function whose graph is symmetrical about the y-axis must be even since $\sin x$ and $\log\left(x + \sqrt{x^2 + 1}\right)$ are odd function.
Therefore $\sin\left(\log\left(x + \sqrt{x^2 + 1}\right)\right)$ must be odd
Also, $\frac{\sec^4 + \operatorname{cosec}^4 x}{x^3 + x^4 \cot x}$ is an odd function
Now, let $f(x) + y = f(x) + f(y) \forall x, y \in R$
 $\therefore f(0+0) = f(0) + f(0) \therefore f(0) = 0$
 $f(x-x) = f(x) + f(-x)$ or $0 = f(x) + f(-1)$
i.e., $f(-x) = -f(x) \therefore f(x)$ is odd.
- $\{x\} > 0$ if $x \notin Z$, $\{x\} = 0$ if $x \in Z$
- no value of T exist such that $f(x+T) = f(x)$
- denominator not equal to zero
- $[x]$, x is defined for all values
- draw graph.
- $\sqrt{f(x)}$ defined $f(x) \geq 0$.
- $|x| + x \neq 0$
- $1 - |x| \geq 0$
- $(x-3)(6-x) > 0$

21. Domain of $\cot x = \mathbb{R} \setminus \{n\pi, n \in \mathbb{Z}\}$
22. $5x \in \mathbb{R}$
23. Domain of $\cot x = \mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$
24. Denominator not equal to zero
25. Denominator $\neq 0$
26. $x \geq 4$ then $f(x) = 1$; $x < 4$ then $f(x) = -1$
27. Range of $\cos x = [-1, 1]$
28. $[x^2 + 1]$ is an integer $\Rightarrow$
 $\sin n\pi = 0 \forall x \in \mathbb{R} \Rightarrow f(x) = 0 \forall x \in \mathbb{R}$
29. If $2 \leq x < 12$ then range of $f = \{10\}$
 if $x \geq 12$ then the range of $f = [10, \infty)$
 if $x \geq 2$ then the range of $f = [10, \infty)$
30. Range of $ax^2 + bx + c$ ($a > 0$) is $\left[\frac{4ac - b^2}{4a}, \infty \right)$
31. $y = \frac{2+x}{2-x} \Rightarrow 2y - yx = 2 + x \Rightarrow x(y+1) = 2y - 2$
 $x = \frac{2y-2}{y+1} \Rightarrow f^{-1}(x) = \frac{2x-2}{x+1}$
 $\therefore$ Range $f = \text{domain } f^{-1} = \mathbb{R} - \{-1\}$
32. $x - [x] \in [0, 1)$
33. substitute domain values
34. Use graph
35. $f: \mathbb{R} \rightarrow \mathbb{S}$ is onto $\Rightarrow \mathbb{S} = \text{Range}$
 $f = [1 - \sqrt{1+3}, 1 + \sqrt{1+3}] = [-1, 3]$
36. $x^2 - 2x + 1 + 5 = (x-1)^2 + 5 \geq 5$
37. $f(x_1) = f(x_2) \Rightarrow x_1 = x_2 \Rightarrow f$ is one-one
 let $f(x) = y \Rightarrow ax + b = y \Rightarrow x = \frac{y-b}{a} \in \mathbb{R}$
 $\forall y \in \mathbb{R} \Rightarrow f$ is onto
38. $f(-1) = f(1) = 1 \Rightarrow f$ is not one-one
 $x^2 > 0 \forall x \in \mathbb{R} \Rightarrow \text{Range of } f \neq \text{codomain}$
39. $\sin x$ is a periodic function with period $2\pi \Rightarrow \sin x$ is a many one function.
40. $f(x) = [x] = x \forall x \in \mathbb{R}$, f is identity function

41. $f(x_1) = f(x_2) \Rightarrow x_1 = x_2 \Rightarrow f$ is one-one
 Let $f(x) = y \Rightarrow 15x + 7 = y \Rightarrow x = \frac{y-7}{15} \in \mathbb{Q}$
 $\forall y \in \mathbb{Q} \Rightarrow f$ is onto
42. $f(x_1) = f(x_2) \Rightarrow x_1^2 = x_2^2 \Rightarrow x_1 = x_2 \Rightarrow f$ is one-one
 Range of $f = (0, \infty)$ not equal to codomain
43. $f(x_1) = f(x_2); \frac{2x_1}{1+x_1} = \frac{2x_2}{1+x_2} \Rightarrow x_1 = x_2$
 $\therefore f$ is one-one, Range $\subset$ Co-Domain
 $\therefore f$ is not onto
44. No. of function from A to $A = n(A)^{n(A)} = 256$
45. The number of surjections from A to B is
 $n^r - {}^n C_1 (n-1)^r + {}^n C_2 (n-2)^r - {}^n C_3 (n-3)^r + \dots + (-1)^{n-1} \cdot {}^n C_{n-1}$ where $n(A) = r$ & $n(B) = n$
46. No. of one-one functions from A to $B = {}^{n(B)}P_{n(A)}$
 $(n(B) \geq n(A))$
47. No. of many-one functions from A to B
 $= n(B)^{n(A)} - {}^{n(B)}P_{n(A)}$
48. If $n(A) < n(B)$ then the no. of surjections from A to $B = 0$
49. $n(A)! = 106!$
50. $5^4 - 0 = 625$
51. $g(f(f(2))) = g(f(5)) = g(11) = 121 + 1 = 122$
 $(f \circ g)(x) = f[g(x)] = f[x - [x]]$
 $= f(\{x\}) = [\{x\}] = 0, \sin 0 \leq \{x\} < 1$
52. $f \circ f(x) = x \Rightarrow f(f(x)) = x$
53. $f(g(x)) = \frac{g(x)}{\sqrt{1-(g(x))^2}} = \frac{\frac{x}{\sqrt{1+x^2}}}{\sqrt{1-\frac{x^2}{1+x^2}}} = x$
54. Substitute the values
55. $x < 1$
56. let $f^{-1}(x) = y \Rightarrow x = f(y) = \log_5 y$

$$\Rightarrow y = f^{-1}(x) = 5^x$$

58. Let $f^{-1}(x) = y \Rightarrow x = f(y) = 10y - 7$

$$\Rightarrow y = f^{-1}(x) = \frac{x+7}{10} = g(x)$$

59. Let $f^{-1}(x) = y \Rightarrow x = f(y) = y + \frac{1}{y}$

$$\Rightarrow y^2 - xy + 1 = 0$$

60. $x - [x] = 0$ for all integral values of x . Therefore, the function is many one-and, therefore, not defined.



LEVEL - II (C.W)

Real valued functions:

1. Let $g(x)$ be a function defined on $[-1, 1]$. If the area of the equilateral triangle with two of its vertices at $(0, 0)$ and $(x, g(x))$ is

$\sqrt{3}/4$, then the function $g(x)$ is

1) $g(x) = \pm\sqrt{1-x^2}$ 2) $g(x) = \sqrt{1-x^2}$

3) $g(x) = -\sqrt{1+x^2}$ 4) $g(x) = \sqrt{1+x^2}$

2. If $f: R \rightarrow R$ is defined by $f(x) = x - [x] - \frac{1}{2}$ for $x \in R$, where $[x]$ is the greatest integer not exceeding x , then $\left\{x \in R: f(x) = \frac{1}{2}\right\} =$

1) Z 2) N 3) ϕ 4) R

3. Suppose $f: [2, 2] \rightarrow R$ is defined by

$$f(x) = \begin{cases} -1 & \text{for } -2 \leq x \leq 0 \\ x-1 & \text{for } 0 \leq x \leq 2 \end{cases}$$

then the $\{x \in (-2, 2): x \leq 0 \text{ and } f(|x|) = x\} =$

1) $\{-1\}$ 2) $\{0\}$ 3) $\{-1/2\}$ 4) ϕ

4. If $|\sin x + \cos x| = |\sin x| + |\cos x|$, then x lies

in 1) 1st quadrant only

2) 1st and 3rd quadrant only

3) 2nd and 4th quadrant only

4) 3rd and 4th quadrant only

Even and odd functions:

5. Let $f(x) = \begin{cases} 4 & x < -1 \\ -4x & -1 \leq x \leq 0 \end{cases}$

If $f(x)$ is an even function on R then the definition of $f(x)$ on $(0, \infty)$ is

1) $f(x) = \begin{cases} 4x & 0 < x \leq 1 \\ 4 & x > 1 \end{cases}$ 2) $f(x) = \begin{cases} 4x & 0 < x \leq 1 \\ -4 & x > 1 \end{cases}$

3) $f(x) = \begin{cases} 4 & 0 < x \leq 1 \\ 4x & x > 1 \end{cases}$ 4) $f(x) = \begin{cases} 4 & x < -1 \\ -4x & -1 \leq x \leq 0 \end{cases}$

6. If $f(x) = \begin{cases} x^2 \sin \frac{\pi x}{2} & |x| < 1 \\ x|x| & |x| \geq 1 \end{cases}$ then $f(x)$ is

1) an even function 2) an odd function
3) a periodic function 4) neither odd nor even

7. $f(x) = \frac{\cos x}{\left[\frac{2x}{\pi}\right] + \frac{1}{2}}$, where x is not an

integral multiple of π and $[\cdot]$ denotes the greatest integer function is

1) an odd function 2) even function
3) neither odd nor even 4) both even and odd

Periodic functions:

8. Which of the following function is not periodic

1) $\frac{2^x}{2^{[x]}}$ 2) $\sin^{-1}(\{x\})$

3) $\sin^{-1}(\sqrt{\cos x})$ 4) $\sin^{-1}(\cos(x^2))$

9. Let $f(x) = nx + n - [nx + n] + \tan \frac{\pi x}{2}$, where

$[x]$ is the greatest integer $\leq x$ and $n \in N$. It is

1) a periodic function of period 1

- 2) a periodic function of period 4
3) not periodic
4) a periodic function of period 2
10. Let $f(x) = x(2-x)$, $0 \leq x \leq 2$. If the definition of f is extended over the set $R - [0, 2]$ by $f(x+2) = f(x)$ then f is a
- 1) periodic function of period 1
2) non periodic function
3) periodic function of period 2
4) periodic function of period $1/2$
11. If f is periodic, g is polynomial function and $f(g(x))$ is periodic and $g(2) = 3, g(4) = 7$ then $g(6)$ is
- 1) 13 2) 15 3) 11 4) 21
- Domain of the function:**
12. The domain of the function $f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is [AIE-2004]
- 1) $[2, 3]$ 2) $[1, 2]$ 3) $[1, 2]$ 4) $[2, 3]$
13. The domain of $f(x) = \sin^{-1} \left\{ \log_3 \left(\frac{x^2}{3} \right) \right\}$ is
- 1) $(-\infty, 3]$ 2) $[3, \infty)$
3) $[-3, -1] \cup [1, 3]$ 4) $(-9, -1) \cup (1, 9)$
14. The domain of $f(x) = \log_x (9-x^2)$ is
- 1) $(-3, 3)$ 2) $(0, \infty)$
3) $(0, 1) \cup (1, \infty)$ 4) $(0, 1) \cup (1, 3)$
15. The domain of $f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$ is [AIE-2003]
- 1) $(1, 2)$ 2) $(-1, 0) \cup (1, 2)$
3) $(1, 2) \cup (2, \infty)$ 4) $(-1, 0) \cup (1, 2) \cup (2, \infty)$
16. The domain of $f(x) = \sqrt{1 - \sqrt{1 - \sqrt{1 - x^2}}}$ is
- 1) $[0, 1]$ 2) $[-1, 1]$ 3) $(-\infty, \infty)$ 4) $(-1, 1)$
17. The domain of $f(x) = \cos(\log x)$ is
- 1) $(-\infty, \infty)$ 2) $(-1, 1)$ 3) $(0, \infty)$ 4) $(1, \infty)$

18. The domain of $f(x) = \cos^{-1} \left(\frac{2}{2 + \sin x} \right)$ contained in $[0, 2\pi]$ is
- 1) $\left[0, \frac{\pi}{2}\right]$ 2) $\left[\frac{\pi}{2}, \pi\right]$ 3) $[0, \pi]$ 4) $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$
19. $\{x \in R : [x - |x|] = 5\} =$ (EAM-05)
- 1) R is the set of real numbers
2) ϕ , the null set
3) $\{x \in R : x < 0\}$ 4) $\{x \in R : x \geq 0\}$
20. The domain of the function defined by $f(x) = {}^{(7-x)}P_{(x-3)}$ is
- 1) $\{3, 7\}$ 2) $\{3, 4, 5, 6, 7\}$
3) $\{3, 4, 5\}$ 4) $\{1, 2, 3, 4\}$
21. $f : N \rightarrow N$ is defined as
- $$f(n) = \begin{cases} 2, n = 3k, & k \in Z \\ 10 - n, n = 3k + 1, & k \in Z \\ 0, n = 3k + 2, & k \in Z \end{cases}$$
- then $\{n \in N : f(n) > 2\} =$ (EAM-04)
- 1) $\{3, 6, 4\}$ 2) $\{1, 4, 7\}$ 3) $\{4, 7\}$ 4) $\{7\}$
22. The domain of the function $f(x) = \frac{1}{\sqrt{\{\sin x\} + \{\sin(\pi + x)\}}}$ where $\{\cdot\}$ denotes the fractional part, is
1. $[0, \pi]$ 2. $(2n+1)\frac{\pi}{2}, n \in Z$
3. $(0, \pi)$ 4. $R - \left\{ \frac{n\pi}{2}, n \in Z \right\}$
- Range of the function:**
23. If $\alpha \in (0, \frac{\pi}{2})$, then $\sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}$ is always greater than or equal to ($x \neq 0, -1$)
- 1) 2 2) 1 3) $2 \tan \alpha$ 4) $2 \sec^2 \alpha$

24. The range of $f(x) = \sin^2 x + \cos^4 x$ is

- 1) $\left[\frac{1}{2}, 1\right]$ 2) $\left[\frac{3}{4}, 1\right]$ 3) $[0, 1]$ 4) $\left[0, \frac{1}{4}\right]$

25. The range of $f(x) = \sin^{-1}\left[\frac{1}{2} + x^2\right]$ is $[\cdot]$ denotes greatest integer function)

- 1) $\left\{-\frac{\pi}{2}, 0, \frac{\pi}{2}\right\}$ 2) $\left\{0, \frac{\pi}{2}\right\}$ 3) $\left\{\frac{\pi}{2}\right\}$ 4) $\{0, \pi\}$

26. If $f: R \rightarrow R$ and $g: R \rightarrow R$ defined by

$$f(x) = |x| \text{ and } g(x) = [x-3] \text{ for}$$

$x \in R$ $[\cdot]$ is denotes greatest integer function

then $\left\{g(f(x)) : \frac{-8}{5} < x < \frac{8}{5}\right\} =$ (EAM-08)

- 1) $\{0, 1\}$ 2) $\{1, 2\}$ 3) $\{-3, -2\}$ 4) $\{2, 3\}$

27. The range of $x^2 + 4y^2 + 9z^2 - 6yz - 3xz - 2xy$ is

- 1) ϕ 2) R 3) $[0, \infty)$ 4) $(-\infty, 0)$

28. The maximum possible domain and the corresponding range of $f(x) = (-1)^x$ are

- 1) $D_f = R, R_f = [-1, 1]$ 2) $D_f = Z, R_f = \{1, -1\}$
3) $D_f = Z, R_f = [-1, 1]$ 4) $D_f = R, R_f = \{-1, 1\}$

29. The range of

$$f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x \text{ is}$$

- 1) $(0, \pi)$ 2) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ 3) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ 4) $\left[0, \frac{3\pi}{4}\right]$

Types of functions:

30. Let $A = [-1, 1] = B$ then which of the following function from A to B is bijective function

- 1) $f(x) = \frac{x}{2}$ 2) $g(x) = |x|$
3) $h(x) = x^2$ 4) $k(x) = \sin \frac{\pi x}{2}$

31. If $f: R \rightarrow C$ is defined by

$$f(x) = e^{2ix} \text{ for } x \in R \text{ then, } f \text{ is (Where } C \text{ denotes the set of all Complex numbers)}$$

- 1) One-one 2) Onto

3) One-one and Onto

4) neither one-one nor Onto

32. A function $f: N \rightarrow Z$ defined by

$$f(n) = \begin{cases} \frac{n-1}{2}, & \text{when 'n' is odd} \\ \frac{-n}{2}, & \text{when 'n' is even} \end{cases}, \text{ is}$$

- 1) one-one but not onto
2) onto but not one-one
3) one-one onto
4) neither one-one nor onto

33. M is the set of all 2×2 real matrices.

$f: M \rightarrow R$ is defined by $f(A) = \det A$ for all A in M then f is

- 1) one-one but not onto 2) onto but not one-one
3) neither one-one nor onto 4) bijective

34. Let $f: R - \{n\} \rightarrow R$ be a function defined

$$\text{by } f(x) = \frac{x-m}{x-n} \text{ such that } m \neq n \text{ then}$$

- 1) f is one one into function
2) f is one one onto function
3) f is many one into function
4) f is many one onto function

$$35. f(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ x, & \text{if } x \text{ is irrational} \end{cases}$$

$$g(x) = \begin{cases} 0, & \text{if } x \text{ is irrational} \\ x, & \text{if } x \text{ is rational} \end{cases} \text{ Then } f - g \text{ is}$$

- 1) one-one and into 2) neither one-one nor onto
3) many one and onto 4) one-one and onto

Composite functions:

36. If $f(x)$ and $g(x)$ are two functions with

$$g(x) = x - \frac{1}{x} \text{ and } fog(x) = x^3 - \frac{1}{x^3}, \text{ then } f(x) =$$

- 1) $x^3 + 3x$ 2) $x^2 - \frac{1}{x^2}$ 3) $1 + \frac{1}{x^2}$ 4) $3x^2 + \frac{3}{x^4}$

37. Let $f(x) = ax + b$ and

$$g(x) = cx + d, a \neq 0, c \neq 0. \text{ Assume}$$

$a = 1, b = 2$. If $(f \circ g)(x) = (g \circ f)(x)$ for all x , what can you say about c and d

- 1) c and d both arbitrary 2) $c = 1, d$ arbitrary
3) c arbitrary, $d = 1$ 4) $c = 1, d = 1$

38. If $f(x) = \sin^2 x$ and the composite functions

$g\{f(x)\} = |\sin x|$, then the function $g(x) =$

- 1) $\sqrt{x-1}$ 2) $\sqrt{x}$ 3) $\sqrt{x+1}$ 4) $-\sqrt{x}$

39. If $f: R \rightarrow R$ and $g: R \rightarrow R$ are given by

$f(x) = |x|$ and $g(x) = [x]$ for each $x \in R$,

then $\{x \in R : g(f(x)) \leq f(g(x))\} =$

- 1) $Z \cup (-\infty, 0)$ 2) $(-\infty, 0)$ 3) Z 4) R

40. Let $g: R \rightarrow R$ be given by $g(x) = 3 + 4x$. If

$g^n(x) = g \circ g \circ \dots \circ g(x)$, and $g^n(x) = A + Bx$ then A and B are

- 1) $2^{n+1} - 1, 2^{n+1}$ 2) $4^n - 1, 4^n$
3) $3^n, 3^n + 1$ 4)

Invertibility & inverse of a function:

41. Let 'f' be an injective function with domain

and range $\{1, 2, 3\}$ such that exactly one of the following statements is correct and the remaining are false

the value of is

- 1) x 2) y 3) z 4) x or z

42. If the function is defined

by then is

- 1) 2)
3) 4)

Functional transformations:

43. If $f(x)$ is a polynomial in $x (> 0)$

satisfying the equation

$f(x) + f(1/x) = f(x) \cdot f(1/x)$ and

, then $f(3) =$

- 1) 26 2) 27 3) 28 4) 29

44. A function satisfy the equation

for all

and $f(y) > 0$, then

- 1) 2)
3) 4)

45. f is a function defined as

and

and $f(1) = 2$ then integral value of a

- 1) 3 2) 0 3) 2 4) 1

46. A real valued function $f(x)$ satisfies the functional equation

where a is a given constant and , is equal to [AIE-2004]

- 1) 2)
3) 4)

LEVEL-II(C.W)-KEY

- 01) 2 02) 3 03) 3 04) 2 05) 1
06) 2 07) 1 08) 4 09) 4 10) 3
11) 3 12) 4 13) 3 14) 4 15) 4
16) 2 17) 3 18) 3 19) 2 20) 3
21) 2 22) 4 23) 3 24) 2 25) 2
26) 3 27) 3 28) 2 29) 2 30) 4
31) 4 32) 3 33) 2 34) 1 35) 4
36) 1 37) 2 38) 2 39) 4 40) 2
41) 2 42) 2 43) 1 44) 4 45) 1
46) 4

LEVEL-II(C.W)-HINTS

1. Side of the triangle with vertices and

_____ is a _____. Area of equilateral triangle, whose side a is _____.

_____. Thus _____.

2.

_____.

_____.

3. By verification,

hence _____.

4. The given relation holds only when $\sin x$ and $\cos x$ have same sign or atleast one of them is zero.

hence _____.

5. If _____ is even,

_____.

_____.

6. Here,

_____, _____.

Let _____. Then

_____ and

_____.

Therefore _____.

_____.

_____.

_____ for all _____. Also, none of the pieces of definition are periodic.

7.

_____.

(as _____ is not an integral multiple of _____)

_____.

_____ is an odd function

8. Period of _____ is 1, period of _____ is 1.

period of _____ is _____ where as

_____ is non-periodic.

9. _____ has the period $1/n$ and _____

has the period _____ i.e, LCM of $1/n, 2$ is 2.

10. Conceptual

11. From the given data _____ must be linear

function. Hence, _____.

Also _____ and _____

solving, _____ and _____

Hence, _____ Then, _____

12. _____ is defined _____ and

_____ and _____

_____ domain = $[2, 3)$

13.

_____.

14. $9 - x^2 > 0, x > 0$ and _____

15. _____ is defined

_____ domain = _____

16. _____

17. $x > 0$

18. _____

19. _____

20. _____

21. Verification

22. _____

Now _____

For _____ to get defined _____

integer _____

Hence, the domain is $\mathbb{R}$ -_____

23. _____

24. _____

25. _____

26. _____

27. _____

28. _____ is defined when _____ is an integer

_____ Domain = $\mathbb{Z}$, range = _____

29. Domain _____

30. _____ f is one-one

31. _____ f is not one - one

Range of f _____ C _____ f is not onto

32. _____

_____ is one-one , onto

33. _____

_____ is not one-one

_____ there exist a matrix _____

such that $f(1) = k$ _____ f is onto

34. Given _____

where _____, Let _____

_____ is one-one

Let _____ such that _____,

_____ is not defined for _____, also _____ is not real.

_____ is not onto function.

If a function is not onto it referred that it is into function. Hence _____ is one-one into function.

35.

Clearly is one-one and onto.

36.

let Thus

37.

and

38.

39.

this is true for 40. Since similarly, we get 41. Suppose $f(x)=1$, then $f(y)=1$, $f(z)=2$ is not an injection.Suppose then A contradictionSuppose then $f(y)=1$, $f(z)=3$, $f(x)=2$ this is true

42.

43. Take $f(x) = 1 - x^n$ 44. Take and verify45. $f(x) = 2^x$

46.

Now **LEVEL - II (H.W)****Real valued functions:**1. If then 1) xy 2) 3) 4) 2. If then1) Only when $m = n$ 2) Only when 3) Only when $m = -n$ 4) For all m and n

Even and odd functions:

3. Let and . Then is
 1) an even function 2) an odd function
 3) neither even nor odd 4) periodic
4. If (where denotes the fractional part of x), is
 1) even function 2) odd function
 3) neither even nor odd 4) constant function
5. **Periodic functions:**
 period of
 where is
 1) n 2) 1 3) 4) 2
6. Let . Then is
 1) a periodic function of period
 2) a periodic function of period
 3) not a periodic function
 4) a periodic function of period
7. The function where is the greatest integer less than or equal to is a
 1) periodic function of indeterminate period
 2) periodic function of period
 3) nonperiodic function
 4) periodic function of period 1
8. If then the period of is
 1) 1 2) 3) 4) no fundamental period
9. If and are periodic functions with period 7 and 11, respectively. Then the period of is
 1) 177 2) 222 3) 433 4) 1155

Domain of the function:

10. The domain of $f(x) = \frac{\text{input}}{\text{input}}$ is
 1) 2)
 3) 4)
11. Domain of
 1) 2)
 3) 4)
12. Domain of
 1) 2) 3) $(0, 3)$ 4) $\mathbb{R}$
13. Domain of
 1) $\mathbb{R}$ 2) $\mathbb{Z}$ 3) 4)
14. The domain of is
 1) 2) 3) 4) $\mathbb{R}$
15. The domain of the function $f(x) = \frac{\text{input}}{\text{input}}$ is
 1) $\mathbb{R}$ 2) $[-2, 2]$ 3) $[-1, 1]$ 4)
16. The domain of is
 1) 2)
 3)
 4)
17. then

$\square =$ (EAM-04)
1) $(-1,3)$ 2) $[-1,3)$ 3) $(-1,3]$ 4) $[-1,3]$

18. The domain of $f(x) = \square$ is
1) $\square$ 2) $\square$ 3) $\square$ 4) $\square$

19. The domain of $f(x) = \square$ is
1) $(-2,2)$ 2) $\square$
3) $\square$ 4) $\square$

20. Let f and g be two real functions given by
 $f = \{(0, 1), (2, 0), (3, -4), (4, 2), (5, 1)\}$
 $g = \{(1, 0), (2, 2), (3, -1), (4, 4), (5, 3)\}$
then the domain of $f.g$ is given by

1) $\square$ 2) $\square$
3) $\square$ 4) $\square$

21. The domain of the function f defined by

$\square$ is equal to

1) $\square$ 2) $\square$
3) $\square$ 4) $\square$

22. Domain of $\square$

1) $\square$ 2) $\square$
3) $\square$ 4) $\square$

23. The set of all real numbers satisfying

$\square$ is
1) $\square$ 2) $\square$ 3) $\square$ 4) $(0,1)$

Range of the function :

24. The range of $f(x) = \square \sin \square$ is
1) $[-1,1]$ 2) $[0, 1]$ 3) $[0,8]$ 4) $[0,4]$

25. The range of $f(x) = \square$ is
1) $\square$ 2) $\square$ 3) $[0, 1]$ 4) $\square$

26. The range of $f(x) = \square$ is

1) $\square$ 2) $\square$
3) $\square$ 4) $\square$

27. If $a^2 + b^2 + c^2 = 1$ then the range of $ab + bc + ca$ is

1) $\square$ 2) $\square$ 3) $\square$ 4) $\square$

28. The range of the function $\square$ is
[AIE-2004]

1) $\square$ 2) $\square$
3) $\square$ 4) $\square$

29. Range of $\square$ is

1. $\square$ 2. $\square$
3. $\square$ 4. $\square$

Types of functions:

30. If $f(x) = \square$ when $2 < x < 3$ is

1) one one function only 2) an onto function only
3) into function 4) identify function

31. $\square$ defined by

$\square$ is

1) one-one but not onto 2) onto but not one-one
3) neither one - one nor onto 4) bijective

32. The function $\square$ defined by

$\square$ is
1) one-one but not onto 2) onto but not one-one
3) neither one - one nor onto 4) bijective

33. $\square$ is defined as $\square$,

$\square$ then f is a constant function when

1) $\square$ 2) $\square$ 3) $\square$ 4) $\square$

34. $\square$ is

1) one-one but not onto 2) one-one onto
3) onto but not one-one 4) neither one-one nor onto

35. Let $f: A \rightarrow B$ be a one-one function and only one of the conditions

(i) f is onto, (ii) f is bijective, (iii) f is surjective is true then the function f is given by the set

- 1) $\{(1,1), (2,2), (3,3), (4,4)\}$ 2) $\{(1,2), (2,3), (3,4), (4,1)\}$
3) $\{(1,1), (2,2), (3,4), (4,3)\}$ 4) $\{(1,2), (2,1), (3,4), (4,3)\}$

36. $f: A \rightarrow B$ where $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4\}$ then f is

- 1) one-one and into 2) many-one and into
3) one-one and onto 4) many-one and onto

Composite function:

37. If $f: A \rightarrow B$ and $g: B \rightarrow C$ then

$g \circ f$ is

- 1) $f \circ g$ 2) $f \circ g$
3) $f \circ g$ 4) $f \circ g$

38. Suppose that $f: A \rightarrow B$ and

$g: B \rightarrow C$ then $g \circ f$ is

- 1) $f \circ g$ 2) $f \circ g$ 3) $f \circ g$ 4) $f \circ g$

39. Let $f: A \rightarrow B$ and

$g: B \rightarrow C$ then for all $x \in A$

- 1) x 2) 1 3) $f(x)$ 4) $g(f(x))$

40. If $f: A \rightarrow B$ and $g: B \rightarrow C$ are defined by

$f(x) = x^2$ and $g(x) = x$ then the values of x such that $g(f(x)) = x$ are.

- 1) 1,2 2) -1,2 3) -1,-2 4) 1,-2

Invertibility & inverse of a functions

41. If $f(x) = 1 + x + x^2 + x^3 + \dots$ for $|x| < 1$ then $f^{-1}(x)$ is

- 1) $\frac{1}{1-x}$ 2) $\frac{1}{1+x}$ 3) $\frac{1}{1-x^2}$ 4) $\frac{1}{1+x^2}$

42. If $f: A \rightarrow B$ then domain of f^{-1} for the existence of inverse of f is (EAM-88)

- 1) $[0, \infty)$ 2) $(-\infty, \infty)$ 3) $(-\infty, \infty)$ 4) $(-\infty, \infty)$

43. Let $f: A \rightarrow B$ Then

f is invertible for $A = B$

- 1) $A = B$ 2) $A \neq B$ 3) $A = B$ 4) $A \neq B$

Functional transformations:

44. If $f(x) = x^2$ then

$f(x+1) = x^2 + 2x + 1$

- 1) -2 2) -1 3) $1/2$ 4) 0

45. If $f(x) = x^2$ and $g(x) = x$ then

- 1) 35 2) 36 3) 37 4) 128

46. If $f(x)$ is a function such that

$f(xy) = f(x) + f(y)$ and $f(2) = 1$ then $f(8)$ is

- 1) x^2 2) 2^x 3) $\log_2 x$ 4) $\log_x 2$

47. If f satisfies the relation

$f(x) = f(y) + f(z)$

and

- 1) 0 2) 1 3) 2 4) 3

LEVEL-II(H.W.)-KEY

- 01) 3 02) 4 03) 3 04) 1 05) 2
06) 3 07) 3 08) 1 09) 4 10) 2
11) 1 12) 1 13) 3 14) 1 15) 4
16) 2 17) 1 18) 4 19) 2 20) 4
21) 1 22) 4 23) 2 24) 3 25) 2
26) 4 27) 4 28) 1 29) 3 30) 4
31) 1 32) 3 33) 3 34) 2 35) 3
36) 3 37) 1 38) 2 39) 2 40) 3

- 41) 4 42) 2 43) 3 44) 4 45) 4
46) 3 47) 1

LEVEL-II(H.W.)-HINTS

1. Given $\frac{1}{x} = \frac{1}{y}$
Let $\frac{1}{x} = a$ and $\frac{1}{y} = b$
Then $a = b$
Substituting the value of x and y in (i), We obtain

$$\frac{1}{x} = \frac{1}{y} \Rightarrow \frac{1}{x} = \frac{1}{y} \Rightarrow \frac{1}{x} = \frac{1}{y}$$

2. Let $\frac{1}{x} = a$ and $\frac{1}{y} = b$, then

$$\frac{1}{x} = a \text{ and } \frac{1}{y} = b$$

Given

$$\frac{1}{x} = a \text{ and } \frac{1}{y} = b$$

$$\frac{1}{x} = a \text{ and } \frac{1}{y} = b$$

for all m, n

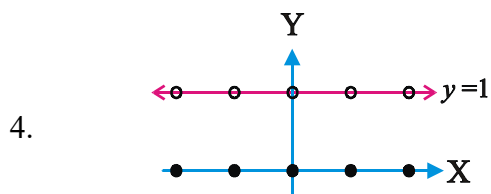
3. $\frac{1}{x} = \frac{1}{y}$

$$\text{If } \frac{1}{x} = \frac{1}{y}$$

$$\text{If } \frac{1}{x} = \frac{1}{y}$$

$$\text{If } \frac{1}{x} = \frac{1}{y}$$

$$\text{If } \frac{1}{x} = \frac{1}{y}$$



5. $\frac{1}{x} = \frac{1}{y}$

$$\frac{1}{x} = \frac{1}{y}$$

period of

6. $\frac{1}{x}$ has the period $\frac{1}{x}$ and $\frac{1}{y}$ has the period $\frac{1}{y}$.

As $\frac{1}{x}$ and $\frac{1}{y}$ do not have a common multiple, $\frac{1}{x}$ is not periodic

7. $\frac{1}{x}$ has the period 1 and $\frac{1}{y}$ has the period $\frac{1}{y}$. Clearly, 1 and $\frac{1}{y}$ do not have a common multiple.

8. Period of $\frac{1}{x}$ is $\frac{1}{x}$.
 $\frac{1}{x}$ period of $\frac{1}{y}$ is half of that of $\frac{1}{x}$.
L.C.M of $\frac{1}{x}$ and $\frac{1}{y}$ is 1].

9. The period of $\frac{1}{x}$ is $\frac{1}{x}$. The period of $\frac{1}{y}$ is $\frac{1}{y}$.

The period of $\frac{1}{x}$ is $\frac{1}{x}$. The period of $\frac{1}{y}$ is $\frac{1}{y}$. Hence, $\frac{1}{x}$ period of $\frac{1}{y}$

$\frac{1}{x}$ and $\frac{1}{y}$

$\frac{1}{x}$ period of $\frac{1}{y}$

$\frac{1}{x}$ period of $\frac{1}{y}$ is 1155.

10. $9 - x^2 > 0$ and $\frac{1}{x}$

11. $\frac{1}{x}$

12. Log is defined for +Ve values.

13. $\frac{1}{x}$

14. and $x > 0$
15. and
16.
17. for $-1 < x < 1$
 $f(x) = [x] > 0$ for
18. and $x - 1 > 0$
19.
20. Common domain of f and g .
21. defined .
22.
23.
24.
25. and x is real.
26. , apply componendo and dividendo
27. If $a^2 + b^2 + c^2 = k$ then
28. We have

- Range of
29. Range
30. $f(x) = x$ when $2 < x < 3$ f is a Identify function (Bijection)
31. Use graph
32. $f(1) = f(-1) = e^1$ f is not one-one
 $e^{|x|} > 0$ $x \in \mathbb{R}$ f is not onto
33. $ad = bc$ or
34. Fact: Every increasing function is one-one onto

 ;

 is an increasing function . So is
 onto function. Also
 so is also one-one.
35.
 (uses (iii) also)
 (uses (ii) also)
 (uses (iii) also)
 (uses (ii) also)

 (uses (i) also)
 (uses (i) also)

 (uses (i) also)

- (uses (i) also)
 (possible)
 (uses (i) also)
36. , when is clearly are one-one and onto
37.
38.
- Put
 Then
 therefore,
39. Here
- (where)
- Now
- Clearly .so,
40.
-
41. Let $f^{-1}(y) = x$

42. is a bijection in
43. By definition of composition of functions

We know is bijection only, when

Thus is bijection if

44. The given expression is

Apply

45. Take $f(x) = k^x$
46. Take $f(x) = \log_a x$
47. If $y = x$ then
if $y = -x$ then



LEVEL - III

1. If for nonzero x ,

where , then $f(2) =$

1)

2)

3)

4)

2. If $\boxed{}$ and a, b are the roots of

$\boxed{}$, then

1)

2)

3)

4)

3. If $\boxed{}$ then

$\boxed{} =$

1) 997 2) 997.5 3) 998 4) 998.5

4. If $[x]$ stands for the greatest integer function, then

1) 498

2) 499

3) 500

4) 501

5. Let $\boxed{}$ $\mathbb{R}$ where

$\boxed{}$ be an odd

function then the value of a is (where $[\cdot]$ represents greatest integer function)

1) less than 11 2) 11 3) greater than 11 4) 12

6. If $\boxed{}$ for all real $\boxed{}$ and $\boxed{}$, then the function

$\boxed{}$ is

1) even function

2) odd function

3) odd if $\boxed{}$

4) neither even nor odd

7. Let $\boxed{}$,

then $\boxed{}$ is (where $[\cdot]$ represents greatest integer function)

1) an odd function

2) an even function

3) both odd and even 4) neither odd nor even

8. If the real valued function $\boxed{}$

is even, then $n =$

1) 2

2) $\boxed{}$ 3) $\boxed{}$

4) 3

9. If f is an even function defined on the interval $\boxed{}$ then find the total number of real values of x satisfying the equations

$\boxed{}$ are

1) 1

2) 2

3) 4

4) 8

10. Let the function $\boxed{}$

be defined on

the interval $[0, 1]$. The even extension of

$\boxed{}$ of the interval $[-1, 1]$ is

1) $\boxed{}$ 2) $\boxed{}$ 3) $\boxed{}$ 4) $\boxed{}$

11. If $\boxed{}$ is an invertible function such that $\boxed{}$ and $\boxed{}$ are symmetric about the line $\boxed{}$ then

1) $\boxed{}$ is odd

2) $\boxed{}$ and $\boxed{}$ may not be symmetric about the line $\boxed{}$

3) $\boxed{}$ may not be odd 4) $\boxed{}$ may be odd

12. If $\boxed{}$ is a function satisfying the property $\boxed{}$,

then the period of $\boxed{}$ is

1) 2

2) 4

3) 8

4) 12

13. If the period of the function $f(x) = \sin\left(\frac{\pi}{n}x\right)$ where $[n]$ denotes the greatest integer less than or equal to n is $\frac{1}{2}$, then

- 1) $\frac{1}{4}$ 2) $\frac{1}{2}$
3) $\frac{3}{4}$ 4) 1

14. The period of the function $f(x) = \sin\left(\frac{\pi}{2}x\right)$ is

- 1) 1 2) 2 3) Non periodic 4) 4

15. The period of the function

$f(x) = \sin\left(\frac{\pi}{2}x\right)$ is

- 1) $\frac{1}{2}$ 2) $\frac{1}{4}$ 3) $\frac{3}{4}$ 4) 1

16. The period of the function

$f(x) = \sin\left(\frac{\pi}{2}x\right)$ is

- 1) $\frac{1}{2}$ 2) $\frac{1}{4}$
3) $\frac{3}{4}$ 4) 1

17. The period of the function

$f(x) = \sin\left(\frac{\pi}{2}x\right)$ is

- 1) $\frac{1}{2}$ 2) $\frac{1}{4}$ 3) $\frac{3}{4}$ 4) 1

18. If $f(x)$ is an odd periodic function with period 2, then $f(4) =$

- 1) $\frac{1}{2}$ 2) 4 3) 2 4) 0

19. If f is periodic g is polynomial function and

$f(x) = \sin\left(\frac{\pi}{2}x\right)$ is periodic and $g(x) = \cos\left(\frac{\pi}{2}x\right)$

then $f(x) + g(x)$ is

- 1) 13 2) 15 3) 11 4) 16

20. The domain of the function $f(x) = \sin\left(\frac{\pi}{2}x\right)$ is

- 1) $\frac{1}{2}$ 2) $\frac{1}{4}$ 3) $\frac{3}{4}$ 4) R

21. The largest interval lying in $[0, 2\pi]$ for

which the function is

$f(x) = \sin\left(\frac{\pi}{2}x\right)$ defined, [AIE-2007]

- 1) $\frac{1}{2}$ 2) $\frac{1}{4}$
3) $\frac{3}{4}$ 4) 1

22. The domain of $f(x) = \sin\left(\frac{\pi}{2}x\right)$ is

- 1) $\frac{1}{2}$ 2) $\frac{1}{4}$ 3) $\frac{3}{4}$ 4) 1

23. The domain of

$f(x) = \sin\left(\frac{\pi}{2}x\right)$ is

- 1) $\frac{1}{2}$ 2) $\frac{1}{4}$
3) $\frac{3}{4}$ 4) 1

24. The domain of $f(x) = \sin\left(\frac{\pi}{2}x\right)$ is

- 1) $\frac{1}{2}$ 2) $\frac{1}{4}$
3) $\frac{3}{4}$ 4) 1

25. If $f(x) = \sin\left(\frac{\pi}{2}x\right)$ and $g(x) = \cos\left(\frac{\pi}{2}x\right)$, then domain of $f(x) + g(x)$ is

- 1) $\frac{1}{2}$
2) $\frac{1}{4}$
3) $\frac{3}{4}$
4) 1

26. Let $f(x) = \sin\left(\frac{\pi}{2}x\right)$ and

$g(x) = \cos\left(\frac{\pi}{2}x\right)$, the domain of $f(x) + g(x)$ is

- 1) 2) 3) 4) $\mathbb{R}$

27. The function

is defined on the set S , then S is equal to

- 1) 2) 3) 4)

28. If then the domain of

is

- 1)
2)
3) 4)

29. If is defined on , then the domain of is

- 1)
2)
3)
4)

30. Domain of

- 1) 2)
3) 4)

31. The domain of is

- 1) 2)
3) 4)

32. If and , then P lies in interval

- 1) 2) 3) 4)

33. The range of is

- 1) 2)
3) 4)

34. The image of the interval under the mapping , given by

is

- 1) 2) 3) 4)

35. The range of is

- 1) 2) 3) 4)

36. If , then range of is

- 1) 2) 3) 4)

37. The range of is

- 1) 2) 3) 4) $\mathbb{R}$

38. The range of is

- 1) 2) 3) 4)

39. If is defined by for , where is the greatest integer not exceeding , then the range of is

1. 2.
3. 4.

40. If are real

constants) and $\lfloor \cdot \rfloor$ then the range

of $\lfloor \cdot \rfloor$ is

- 1) $\mathbb{R}$ 2) $\mathbb{Q}$
3) $\mathbb{Z}$ 4) $\mathbb{N}$

41. The range of $\lfloor \cdot \rfloor$

where $\lfloor \cdot \rfloor$ denotes the greatest integer function, is

- 1) $\mathbb{R}$ 2) $\mathbb{Q}$ 3) $\mathbb{Z}$ 4) $\mathbb{N}$

42. The range of

$\lfloor \cdot \rfloor$,
 $\lfloor \cdot \rfloor$

where $\lfloor \cdot \rfloor$ denotes the greatest integer function $\lfloor \cdot \rfloor$ is

- 1) $\mathbb{R}$ 2) $\mathbb{Q}$ 3) $\mathbb{Z}$ 4) $\mathbb{N}$

43. The function $f(x) = \lfloor x \rfloor$ is defined by

$f(x) = \lfloor x \rfloor$ where $\lfloor \cdot \rfloor$ is G.I.F is surjective then $B =$

- 1) $\mathbb{R}$ 2) $\mathbb{Q}$ 3) $\mathbb{Z}$ 4) $\mathbb{N}$

44. If $f(x) = \lfloor x \rfloor$, $g(x) = \lfloor x \rfloor$

and $h(x) = \lfloor x \rfloor$ then the mapping

f, g, h is

- 1) One - one but not onto
2) Onto but not one - one
3) Both one - one and onto
4) Neither one - one nor onto

45. The functions $f(x) = \lfloor x \rfloor$ is defined by

$f(x) = \lfloor x \rfloor$ is

- 1) One one onto 2) One one into
3) Many one onto 4) Many one into

46. Let S be the set of all triangles and $\mathbb{R}^+$ be the set of positive real numbers. Then the function $f: S \rightarrow \mathbb{R}^+$ where

$f(T) = \text{Area}(T)$ is

- 1) injective but not surjective
2) surjective but not injective
3) injective as well as surjective
4) neither injective nor surjective

47. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$f(x) = \lfloor x \rfloor$ then f is

- 1) one-one onto 2) one-one into
3) many one onto 4) many one into

48. If $f(x) = \lfloor x \rfloor$ then $f(x)$

- 1) $\lfloor x \rfloor$ 2) $\lfloor x \rfloor$
3) $\lfloor x \rfloor$ 4) Cannot say

49. If $f(x) = \lfloor x \rfloor$

$g(x) = \lfloor x \rfloor$ and $h(x) = \lfloor x \rfloor$

then f, g, h

- 1) 1 2) 0 3) $\sin x$ 4) $\cos x$

50. If $f(x) = \lfloor x \rfloor$ and

$g(x) = \lfloor x \rfloor$ then which is $f \circ g$

- 1) $\lfloor x \rfloor$ 2) $\lfloor x \rfloor$ 3) $\lfloor x \rfloor$ 4) $\lfloor x \rfloor$

51. If $f(x) = \lfloor x \rfloor$ is a function defined by

$f(x) = \lfloor x \rfloor$ where $\lfloor \cdot \rfloor$ is G.I.F then

- 1) $\lfloor x \rfloor$ 2) $\lfloor x \rfloor$ 3) $\lfloor x \rfloor$ 4) $\lfloor x \rfloor$

52. If the function $f(x)$ is defined by $f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ x+1 & \text{if } x < 0 \end{cases}$ then $f(f(x)) =$

- 1) $x^2 + 1$ 2) $x^2 + x + 1$
3) $x^2 + 2x + 1$ 4) $x^2 + 3x + 1$

53. Let $f(x) = x^2 - 1$ then the solution of the equation $f(x) = 0$ is

- 1) $x = 1$ 2) $x = -1$ 3) $x = 0$ 4) $x = 0$

54. Let $f(x) = x^2 - 1$ be given by

- 1) $f(x) = x^2 - 1$ 2) $f(x) = x^2 + 1$
3) does not exist because f is not one-one
4) does not exist because f is not onto

55. The inverse of the function

- $f(x) = x^2 - 1$ is given by
1) $f^{-1}(x) = \sqrt{x+1}$ 2) $f^{-1}(x) = -\sqrt{x+1}$
3) $f^{-1}(x) = \pm\sqrt{x+1}$ 4) $f^{-1}(x) = \sqrt{x+1}$

56. If $f(x) = x^2 - 1$ and

- $f(f(x)) = 0$ then $x =$
1) a 2) b 3) ab 4) $3ab$

LEVEL-III-KEY

- 01) 2 02) 3 03) 2 04) 3 05) 3
06) 1 07) 1 08) 4 09) 3 10) 1
11) 1 12) 2 13) 1 14) 3 15) 3
16) 1 17) 2 18) 4 19) 3 20) 1
21) 4 22) 2 23) 3 24) 2 25) 4
26) 3 27) 1 28) 3 29) 1 30) 3
31) 4 32) 1 33) 2 34) 2 35) 1
36) 2 37) 1 38) 2 39) 2 40) 1

- 41) 2 42) 3 43) 4 44) 3 45) 4
46) 2 47) 1 48) 1 49) 1 50) 3
51) 2 52) 2 53) 1 54) 1 55) 4
56) 3

LEVEL-III-HINTS

- Replace x by $\frac{1}{x}$ then solve.
- We have, $f(x) = x^2 - 1$
 $f(f(x)) = 0$
 $f(x^2 - 1) = 0$
 $(x^2 - 1)^2 - 1 = 0$
 $x^4 - 2x^2 + 1 - 1 = 0$
 $x^4 - 2x^2 = 0$
 $x^2(x^2 - 2) = 0$
 $x^2 = 0$ or $x^2 = 2$
 $x = 0$ or $x = \pm\sqrt{2}$
[since a, b all roots of $x^4 - 2x^2 = 0$, $x = 0, \pm\sqrt{2}$]
Similarly $f^{-1}(x) = \pm\sqrt{x+1}$
- $f(x) = x^2 - 1$
- if $f(x) = x^2 - 1$
GE $f(f(x)) = 0$
 $= 500$
- $f(x) = x^2 - 1$
 $f(f(x)) = 0$
Now $f(x) = x^2 - 1$ given
 $f(f(x)) = 0$
 $(x^2 - 1)^2 - 1 = 0$
 $x^4 - 2x^2 + 1 - 1 = 0$
 $x^4 - 2x^2 = 0$
 $x^2(x^2 - 2) = 0$
 $x^2 = 0$ or $x^2 = 2$
 $x = 0$ or $x = \pm\sqrt{2}$
- Given $f(x) = x^2 - 1$. Put $x = \frac{1}{x}$,
then $f\left(\frac{1}{x}\right) = \frac{1}{x^2} - 1$.
Put $\frac{1}{x} = x$, then
 $f(x) = x^2 - 1$

Now,

7. The denominator is

8.

9.

10.

11. Since $\sin x$ and $\cos x$ are symmetric about the line $x = \frac{\pi}{4}$

If $\sin x$ lies on $\cos x$ then $\cos x$ lie on $\sin x$

$\sin x$ lies on $\cos x$ is odd.

12. We have

Replace x by $x + 1$,

Replace x by $x + 2$

From (1) and (3)

We get

i.e.,

$\sin x$ is periodic with period $2k$.

13. $\sin x$ is a periodic function with period 2π , therefore $\sin x$ is a periodic function

with period 2π

But the period of $f(x)$ is 2π (given)

14. Let

Then, $\sin nx$ where n changes with x clearly no constant $k > 0$ is possible for which $\sin nx$ corresponding to all x .

$\sin nx$ is a non periodic function.

15. We have

for all x

$\sin x$ is periodic with period 2π .

16. Since $\sin x$ and $\cos x$ are periodic function with

period .

and period of $\sin$ is

of

17. Since is a periodic function with period ,

is a periodic function

with period

18. ;

19. From the given data must be linear function. Hence,

Also and

Solving, we get and

Hence, Then, .

20. is defined for $a, b > 0$ and

21. is defined if

From (1) and (2):

22. is defined if

Domain of

23. is defined then

24. is defined if

25. We have,

26. We have domain of f is

For the function

domain of is

domain of

27. For the two components to be meaningful, we must have

Hence,

28.

$= \log$

since $a > 0$ is defined if and

29. Since the domain of f is $(0,1]$

30.

31. For $\square$ to be defined, we must have

Also,

Now,

Also

Thus,

32. We have,

Also,

33.

is defined if $\square$: $\square$ Range of $\square$

34. Since the given function has minimum value 75 which is attained at $x = 2$ and maximum value is 89 which is attained at $x = 3$ Hence the range of f is $[75, 89]$

35. If $x > 0$ then $f(x) = 0$ If $x < 0$ then $f(x) = \tan hx$ If $\square$ then $\square$

Range does not contain -1

Range = $(-1, 0]$

36.

When $\square$ is rational say $\square$, then $\square$ is a multiple of $\square$ and $\square$ When $\square$ is irrational, $\square$ multiple of $\square$ Thus $\square$ when $\square$ is irrational
= 1, when $\square$ is rational.Range $\square$

37.

38.

39.

or

or

or

40.

which has

the range $\square$

41.

Thus, from domain point of view,

or

42.

where

(as p is an integer)

Now, for

The range of is 1.

43. verify by taking integer & decimal number.

44. Let , then

which is bijective in

Hence is bijective

45. Clearly from the graph, f is many one into function

46. Two triangles may have equal areas

is not one-one.

Since each positive real number can represent area of triangle.

os onto.

47.

since -1

let

since and

is an increasing function therefore said to be 1-1 function

Now and

is continuous also

Range of codomain of

48.

(impossible)

49.

50.

or

51.

52. Let $f(x)=y$

53.

54. Let $y = f(x)$ 55. Let $y = f(x)$

56. Put . Then



LEVEL - IV

Assertion & Reason Type Q - 1 to Q - 5

a) Assertion is True, Reason is True, Reason is a correct explanation for Assertion

b) Assertion is True, Reason is True, Reason is not a correct explanation for Assertion

c) Assertion is True, Reason is False.

d) Assertion is False, Reason is True.

1. **Assertion :** The domain of

, is

Reason : The domain of is $(0, 4)$.

1) a 2) b 3) c 4) d

2. **Assertion :** Every function can be uniquely expressed as the sum of an even function and an odd function.**Reason :** The set of values of parameter 'a' for which the function $f(x)$ defined ason the set $[-3, 3]$ is

an odd function is, .

1) a 2) b 3) c 4) d

3. **Assertion :** The domain of the function

is .

Reason : If , then

cannot attain the value .

1) a 2) b 3) c 4) d

4. **Assertion :** The function $f(x)$ given by

is an odd function.

Reason : The composite function of two odd functions is an odd function

1) a 2) b 3) c 4) d

5. **Assertion :** If $\{x\}$ and $[x]$ represent fractional part and integral part of x then

.

Reason : and , where I is integer

1) a 2) b 3) c 4) d

Statements

6. **Statement I:** The product of two odd functions is an even function.**Statement II:** A constant function is always a bijection.

1) only I is true 2) only II is true

3) both I and II are true 4) neither I nor II true

7. **Statement I :** Every strictly monotonic function is one one**Statement II :** The function defined by is one one.

1) only I is true 2) only II is true

3) both I & II are true

- 2) A - iii; B - iv; C - ii; D - v
3) A - iv; B - iii; C - ii; D - i
4) A - iii; B - vi; C - v; D - ii
12. Match the following
- | List - I | List - II |
|---|-------------------------------------|
| I. $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = ax + b$ is | a) injection but not surjection |
| II. $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = [x]$ is | b) surjection but not injection |
| III. $f: \mathbb{R} \rightarrow [0, \infty)$ defined by $f(x) = x $ is | c) bijection |
| IV. $f: \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^3$ is | d) neither injection nor surjection |
| 1) c, d, a, b | 2) c, d, b, a |
| 3) d, c, b, a | 4) d, c, a, b |
13. If f, g, h, ϕ are functions from $\mathbb{R} \rightarrow \mathbb{R}$ then
- | List - I | List - II |
|---|----------------------------------|
| I. $f(x) = \sin^3 x + 2 \sin x + \sin 3x$ | a) even function |
| II. $g(x) = \cos 3x + \sin^4 x + 3 \cos x$ | b) odd function |
| III. $h(x) = \sin 3x + \cos 2x + \sin^2 x$ | c) both even and odd function |
| IV. $\phi(x) = \frac{\tan[x]\pi}{1 + \sin[x]\pi}$ | d) neither even nor odd function |
| 1) I-b; II-a; III-c; IV-d | 2) I-a; II-b; III-d; IV-c |
| 3) I-a; II-b; III-c; IV-d | 4) I-b; II-a; III-d; IV-c |
14. let $\mathbb{R}$ denote the set of all real numbers and $\mathbb{R}^+$ denote the set of all positive real numbers. For the subsets A and B of $\mathbb{R}$ define $f: A \rightarrow B$ by $f(x) = x^2$ for $x \in A$. Observe the two lists given below.
- | List - I | List - II |
|--|---------------------------------------|
| I. f is one-one and onto if | a) $A = \mathbb{R}^+, B = \mathbb{R}$ |
| II. f is one-one but not onto if | b) $A = B = \mathbb{R}$ |
| III. f is onto but not one-one if | c) $A = \mathbb{R}, B = \mathbb{R}^+$ |
| IV. f is neither one-one nor onto if | d) $A = B = \mathbb{R}^+$ |
- The correct matching of List I to List II is
- | | |
|---------------------------|---------------------------|
| 1) I-a; II-b; III-c; IV-d | 2) I-d; II-b; III-a; IV-c |
| 3) I-d; II-a; III-c; IV-b | 4) I-d; II-b; III-c; IV-a |

15. Observe the following lists

List - I

Functions

List - II

Domain

I. $f(x) = \sqrt{\sin^{-1}(\log_2 x)}$ a) $\{-1, 1\}$

II. $g(x) = \sqrt{\cos(\sin x)}$ b) $\{1\}$

III. $h(x) = \sin^{-1}\left(\frac{1+x^2}{2x}\right)$ c) $\{-1\}$

IV. $f(x) = g(x) + h(x)$ d) $[1, 2]$
e) $\mathbb{R}$

The correct matching of List I to List II is

1) I-b; II-a; III-c; IV-d

2) I-a; II-b; III-d; IV-c

3) I-d; II-e; III-a; IV-a

4) I-b; II-a; III-d; IV-c

LEVEL-IV-KEY

01) 3 02) 4 03) 1 04) 1 05) 1

06) 1 07) 3 08) 3 09) 3 10) 3

11) 1 12) 2 13) 4 14) 3 15) 3

LEVEL-IV-HINTS

1. $\cos^{-1} x$ is defined for $-1 \leq x \leq 1$
2. $f(-x) = -f(x)$
3. Denominator $\neq 0$
4. $f(-x) = -f(x)$
7. $f'(x) > 0$, increasing on $\mathbb{R}^+$
9. $\sin n\pi = 0$, $n \in \mathbb{Z}$
10. Choose numerical values, verify
11. Substitute values
15. $\sqrt{f(x)}$ is defined for $f(x)$ has to be defined and $f(x) \geq 0$